PHYSICS

1) If a physical quantity has dimensional formula $[ML^2T^{-2}A^{-1}]$ and during measurement percentage error in M, L, T, A are 1%, 2%, $\frac{1}{2}$% and 1% then maximum percentage error in physical quantity will be -

(1) 8%

(2) 10%

(3) 9%

(4) 7%

2) The speed at the maximum height of a projectile is $\frac{1}{3}$ rd of its initial speed u. If Its range on the horizontal plane is $\frac{n\sqrt{2}u^2}{9g}$ then value of 5n is

(1) 10

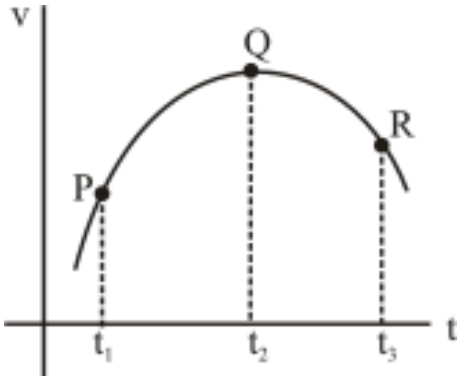
(2) 30

(3) 20

(4) 40

3)

Velocity-time graph of an object moving along x-axis is given in the figure. Point Q is maxima. Select incorrect option :-



(1) At time t_1 , $\vec{a}.\vec{v}$ is positive

(2) At time t_3, acceleration and velocity vector are antiparallel.

(3) At time t_2, object is speeding up

(4) At time t_1, object is speeding up

4) If the radius of curvature of the path of two particles of same mass are in the ratio 3 : 4, then in order to have same constant centripetal force, their velocities will be in the ratio of :

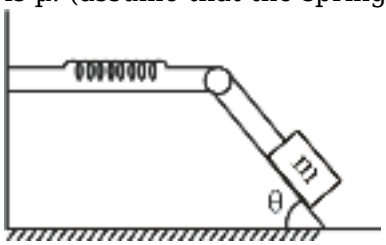
(1) $\sqrt{3} : 2$

(2) $1 : \sqrt{3}$

(3) $\sqrt{3} : 1$

(4) $2 : \sqrt{3}$

5) A block of mass m is situated on a rough incline. It is connected to a spring of spring constant k. The block is released from rest with the spring in the unstretched position. The block moves distance x down the incline before coming to rest. The coefficient of friction between the block and the incline is μ. (assume that the spring has a negligible mass and pulley is frictionless).



Choose the correct relation.

(1) $\frac{1}{2}kx^2 + \mu mgx\cos\theta = mgx\sin\theta$

(2) $\frac{1}{2}kx^2 + mgx\sin\theta = \mu mgx\cos\theta$

(3) $\frac{1}{2}kx^2 = \mu mgx\cos\theta + mgx\sin\theta$

(4) $mgx\sin\theta + \mu mgx\cos\theta + \frac{1}{2}kx^2 = 0$

6) Three photo diodes D_1, D_2 and D_3 are made of semiconductors having band gap of 2eV, 2.5 eV and 3eV respectively which one will be able to detect light of wave length 6000 Å :

(1) D_1

(2) D_2

(3) D_3

(4) D_1 and D_2 both

7) In a Young's double slit experiment, the separation between the slits is 0.15 mm. In the experiment, a source of light of wavelength 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is:

(1) 6.9 mm

(2) 5.9 mm

(3) 4.9 mm

(4) 3.9 mm

8) A monochromatic light is used in Young's double slit experiment when one of the slits is covered by a transparent sheet of thickness 1.8 mm, made of material of refractive index μ_1 number of fringes which shift is 18. when another sheet of thickness 3.6 mm, made of material of refractive index μ_2 is used, number of fringes which shift is 9. Relation between μ_1 and μ_2 is given by.

(1) $4\mu_2 - \mu_1 = 3$

(2) $4\mu_1 - \mu_2 = 3$

(3) $3\mu_2 - \mu_1 = 4$

(4) $2\mu_1 - \mu_2 = 4$

9) **Assertion (A) :** At saturation current only most energetic electrons are reaching anode.
Reason (R) : All emitted photo electrons have same kinetic energy when monochromatic light is used.

(1) Both (A) & (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) & (R) are correct but (R) is not correct explanation of (A)

(3) Both (A) & (R) incorrect

(4) (A) incorrect while (R) correct

10) Assuming photoemission to take place the factor by which the maximum velocity of emitted photo electrons changes when the wavelength of the incident radiation is increased four times is [assume negligible work function]

(1) 4

(2) $\frac{1}{4}$

(3) $\frac{1}{2}$

(4) $\frac{3}{2}$

11) A nucleus at rest splits into two parts having speeds in ratio 1 : 27, then, find the ratio of radius of nucleus.

(1) 2 : 1

(2) 1 : 2

(3) 3 : 1

(4) 1 : 3

12) The radius of the innermost electron orbit of a hydrogen atom is 5.3×10^{-11} m. What is the radius of n = 3 orbit :-

(1) 36.6×10^{-11} m

(2) 47.7×10^{-11} m

(3) 24.2×10^{-11} m

(4) 7.8×10^{-11} m

13) A coil of one loop is made by a wire of length L and there after a coil of two loops is made by same wire. The ratio of magnetic field at the centre of the coil respectively : (current is same in both cases)

(1) 1 : 4

(2) 1 : 1

(3) 1 : 8

(4) 4 : 1

14) A resonance circuit having inductance and resistance 2×10^{-4} H and 6.28 Ω respectively oscillates at 10 MHz frequency. The value of quality factor of this resonator is _____. [π = 3.14]

(1) 200

(2) 150

(3) 2000

(4) 1500

15) **Assertion :** Conduction and displacement current both may be present in same region of space.
Reason : There is no perfectly conducting or perfectly insulating medium exist.

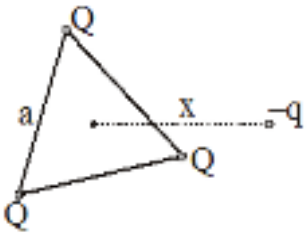
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

16) Three charged particles each of charge +Q are fixed at the corners of an equilateral triangle of side 'a'. A fourth particle of charge –q and mass m is placed at a point on the line passing through centroid of triangle and perpendicular to the plane of triangle at a distance x from the centre of



triangle. Value of x for which the force is maximum is.

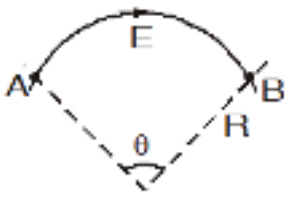
(1) $\frac{a}{\sqrt{3}}$

(2) $\frac{a}{\sqrt{2}}$

(3) $\frac{a}{\sqrt{6}}$

(4) $\frac{a}{\sqrt{5}}$

17) Figure shows an electric line of force which curves along a circular arc. The magnitude of electric field intensity is same at all points on this curve and is equal to E. If the potential at A is V,



then the potential at B is :

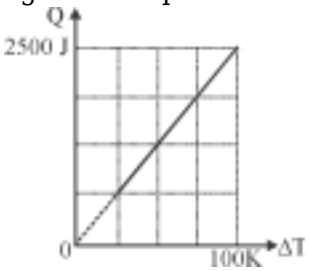
(1) V – ERθ

(2) $V - E2R\sin\frac{\theta}{2}$

(3) V + ERθ

(4) V + 2ER sin $\frac{\theta}{2}$

18) One mole of a gas mixture is heated under constant pressure, and heat required Q is plotted against temperature difference acquired. Find the value of γ for mixture :-



(1) 3/4

(2) 1/4

(3) 3/2

(4) 2/3

19) A liquid of mass 2m and specific heat $\frac{C}{2}$ is heated to a temperature T. Another liquid of mass $\frac{m}{3}$ and specific heat 2C is heated to a temperature 3T. If these two liquids are mixed, the resulting temperature of the mixture is:

(1) $\frac{3T}{5}$

(2) $\frac{9T}{5}$

(3) $\frac{5T}{3}$

(4) $\frac{2T}{3}$

20) A cube of side a and of material of specific gravity s is oscillating on surface of water, vertically up & down the time period of its oscillation is -

(1) $2\pi\sqrt{\frac{a}{g}}$

(2) $2\pi\sqrt{\frac{as}{g}}$

(3) $2\pi\sqrt{\frac{g}{a}}$

(4) $2\pi\sqrt{\frac{g}{sa}}$

21) A simple pendulum is executing simple harmonic motion with a time period T. If the length of the pendulum is increased by 21%, the increase in the time period of the pendulum of increased length is

(1) 10%

(2) 21%

(3) 30%

(4) 50%

22) A steel wire of length 1 m and mass 0.1 kg and having a uniform cross-sectional area of 10^{-6} m^2 is rigidly fixed at both ends. The temperature of the wire is lowered by 20°C. If the wire is vibrating in fundamental mode, find the frequency (in Hz).
($Y_{steel} = 2 \times 10^{11}$ N/m^2, $\alpha_{steel} = 1.21 \times 10^{-5}$/°C)

(1) 11

(2) 20

(3) 15

(4) 10

23) A material has poisson's ratio 0.5. If a uniform rod of it suffers a longitudinal strain of 2×10^{-3}. What is the percentage increase in volume :

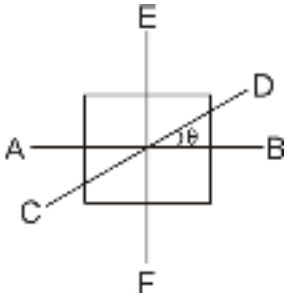
(1) 2%

(2) 4%

(3) 0%

(4) 5%

24) Let I be the moment of inertia of a uniform square plate about an axis AB that passes through its centre and is parallel to two of its sides. CD is a line in the plane of the plate that passes through the centre of the plate and makes an angle θ with AB as shown in figure. The moment of inertia of the plate about the axis CD is then equal to



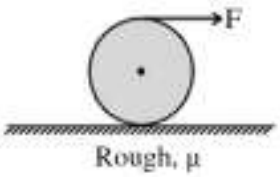
(1) I

(2) I $\sin^2 \theta$

(3) I $\cos^2 \theta$

(4) I $\cos^2 (\theta/2)$

25) A solid sphere of mass M and radius R is pulled by a force F as shown in figure. If the sphere



does not slip over the surface, then frictional force acting on the sphere is

(1) $\frac{2F}{7}$

(2) $\frac{3F}{7}$

(3) $\frac{2F}{5}$

(4) Zero

26) A uniform rod of mass m and length L rotates in horizontal plane with an angular velocity ω about a vertical axis passing through one end. The tension in the rod at a distance x from the axis is :-

(1) $\frac{1}{2}m\omega^2x$

(2) $\frac{1}{2}m\omega^2\left(\frac{x^2}{L}\right)$

(3) $\frac{1}{2}m\omega^2L\left(1-\frac{x}{L}\right)$

(4) $\frac{1}{2}\frac{m\omega^2}{L}\left(L^2-x^2\right)$

27) Consider a wire having current 10A having area of crossection $1cm^2$. If number of electrons per unit volume is 9×10^{28} m^{-3}. Find the drift velocity of electrons :-

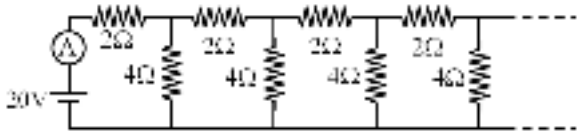
(1) 6.94×10^{-6} m/s

(2) 1.94×10^{-3} m/s

(3) 5.94×10^{-2} m/s

(4) 2.94×10^{-4} m/s

28) An infinite ladder network of resistance is shown in figure, find reading of ammeter-



(1) 2 A

(2) 3 A

(3) 4 A

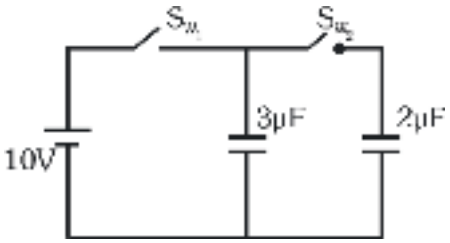
(4) 5 A

29)

A galvanometer has a coil of resistance 100 Ω showing a full–scale deflection at 50 μA. Consider following statements.
(A) The resistance needed to use it as a voltmeter of range 50 volt is 10^6 Ω
(B) The resistance needed to use it as a voltmeter of range 50 volt is 10^5 Ω
(C) The resistance needed to use it as an ammeter of range 10 mA is 0.5 Ω
(D) The resistance needed to use it as an ammeter of range 10 mA is 1.0 Ω

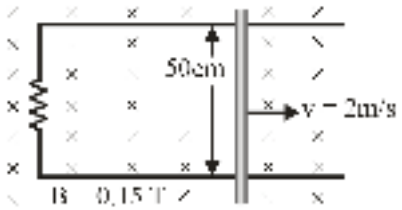
Select **correct** alternative :-
(1) Only A, D
(2) Only A, C
(3) Only B, D
(4) Only B, C

30) In given circuit switch S_{W1} is closed and S_{W2} is open. After long time S_{W1} is opened and S_{W2} is closed. Calculate charge on each capacitor.



(1) 18mC, 12mC
(2) 15mC, 15mC
(3) 10mC, 30mC
(4) 5mC, 25mC

31) As shown in the figure a metal rod makes contact and completes the circuit. The circuit is perpendicular to the magnetic field with B = 0.15 tesla. If the resistance is 3Ω, force needed to move the rod as indicated with a constant speed of 2m/sec is :



(1) 3.75×10^{-3} N
(2) 3.75×10^{-2} N
(3) 3.75×10^{2} N
(4) 3.75×10^{-4} N

32) A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be :

(1) 4 A/m
(2) 0.75 A/m

(3) 2.25 A/m

(4) 1 A/m

33) The rms value of electric field of the light coming from sun is 720 N/C. The average total energy density of the electromagnetic wave is :-

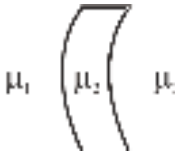
(1) 3.3×10^{-3} J/m^3

(2) 4.58×10^{-6} J/m^3

(3) 6.37×10^{-9} J/m^3

(4) 81.35×10^{-12} J/m^3

34) Focal length of the lens is nR as shown in the figure. If the radii of curvature of the both the surface are equal to R, then value of n is ($\mu_1 = 1$, $\mu_2 = 0.5$, $\mu_3 = 2$) if an object placed at infinity:-



(1) 1

(2) 2

(3) 1.5

(4) 2.5

35) In a compound microscope, the focal length of the objective and the eye lens are 2.5 *cm* and 5 *cm* respectively. An object is placed at 3.75 *cm* before the objective and image is formed at the least distance of distinct vision, then the distance between two lenses will be (i.e. length of the microscopic tube)

(1) 11.67 *cm*

(2) 12.67 *cm*

(3) 13.00 *cm*

(4) 12.00 *cm*

36)

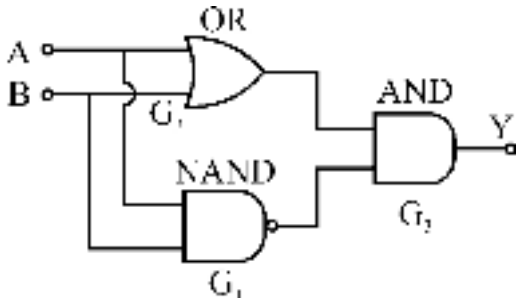
A light ray incident on interface of two media, a to b. If angle of incidence in a is $\theta_a = 45°$ & angle of refraction in medium b is $\theta_b = 60°$. If velocity of light in medium b is $\frac{c}{3}$ then find velocity of light in medium a :-

(1) $\frac{c}{3}\sqrt{\frac{2}{3}}$

(2) $\frac{c\sqrt{2}}{3}$

(3) $\frac{c}{\sqrt{3}}$

(4) None



37) The following configuration of gate is equivalent to :-

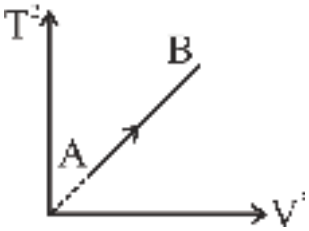
(1) NAND

(2) XOR

(3) OR

(4) None of these

38) An ideal diatomic gas follows the process, as shown in graph where T(K) is temperature and V is volume (m^3). The molar heat capacity of this process will be (in terms of gas constant R):



(1) $\frac{7R}{2}$

(2) 5R

(3) $\frac{19R}{6}$

(4) $\frac{11R}{2}$

39) The power emitted per unit area of a black body is R watt/m^2. At what wavelength will the power radiated by the black body be maximum, If the stefan's constant is σ and Wein's constant is b, then:-

(1) $\lambda = \left(\frac{b\sigma}{R}\right)^{1/4}$

(2) $\lambda = \left(\frac{b}{R}\right)^{1/4}$

(3) $\lambda = \left(\frac{bR}{\sigma}\right)^{1/4}$

(4) $\lambda = b\left(\frac{\sigma}{R}\right)^{1/4}$

40) Volume of liquid flowing per second through a capillary tube is given by $V = \frac{K[x_2 - x_1]^4}{[y_2 - y_1]}$ where x_1 and x_2 are inner and outer radii and y_2 and y_1 are positions crossed by the liquid. The dimensional formula for K is :-

(1) $M^0L^0T^0$

(2) MLT^2

(3) $ML^{-1}T^{-1}$

(4) $M^0L^0T^{-1}$

41) In vernier callipers 1 cm on main scale is divided in 10 equal parts. If 10 division of vernier scale coincides with 8 division of main scale then least count will be :-

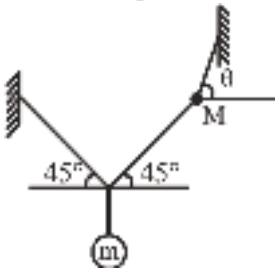
(1) 0.05 cm

(2) 0.01 cm

(3) 0.02 cm

(4) 0.005 cm

42) For equilibrium



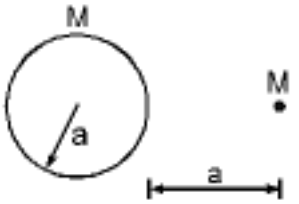
(1) $\tan\theta = 1 + \frac{2M}{m}$

(2) $\tan\theta = 1 + \frac{3M}{m}$

(3) $\tan\theta = 1 + \frac{M}{2m}$

(4) $\tan\theta = 1 + \frac{m}{2M}$

43) A particle of mass M is at a distance 'a' from surface of a thin spherical shell of uniform equal mass and having radius a, Then



(1) Gravitational field & potential both are zero at centre of the shell

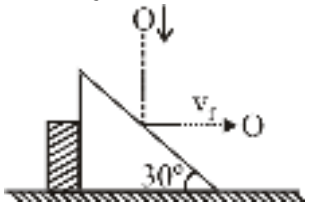
(2) Gravitational field is zero at inside as well as outside points

(3) Inside the shell, gravitational field alone is zero

(4) Neither gravitational field nor gravitational potential is zero inside the shell.

44) As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If ∠ of inclined is 30°, the

velocity v_f will be :-



(1) 3m/s

(2) $\sqrt{3}$ m/s

(3) $1/\sqrt{3}$ m/s

(4) this is not possible

45) Two metal spheres are falling through a liquid of density 2×10^3 kg/m^3 with the same uniform speed. The material density of sphere 1 and sphere 2 are 8×10^3 kg/m^3 and 11×10^3 kg/m^3 respectively. The ratio of their radii is :

(1) $\frac{11}{8}$

(2) $\sqrt{\frac{11}{8}}$

(3) $\frac{3}{2}$

(4) $\sqrt{\frac{3}{2}}$

CHEMISTRY

1) The number of molecules of water in 3.33g of $Al_2(SO_4)_3.18H_2O$ is - (Given Al = 27, S = 32, O = 16, H = 1)

(1) $18.0 \times 6.02 \times 10^{23}$

(2) $9.0 \times 6.02 \times 10^{23}$

(3) $9.0 \times 6.02 \times 10^{21}$

(4) $36.0 \times 6.02 \times 10^{20}$

2) Given that the mass of electron is 9.1×10^{-31} kg and velocity of electron is 2.2×10^6 ms^{-1}. If the uncertainty in its velocity is 0.1%, the uncertainty in position would be :-

(1) 26 nm

(2) 32 nm

(3) 48 nm

(4) 50 nm

3) The energy of second Bohr's orbit in H-atom is –328 kJ mol^{-1}, hence the energy of fourth Bohr's orbit would be ?

(1) –82.00 kJ mol^{-1}

(2) –41 kJ mol^{-1}

(3) –1312 kJ mol^{-1}

(4) –164 kJ mol^{-1}

4) $(NH_4)_2CO_{3(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)} + H_2O_{(g)}$
The value of K_p for above equilibrium is 64 atm^4. The equilibrium pressure of the system if only $(NH_4)_2CO_3$ is present initially is :

(1) 4 atm

(2) 8 atm

(3) 2 atm

(4) 6 atm

5)

For the reaction,
$4NH_{3(g)} + 5O_{2(g)} \rightleftharpoons 4NO_{(g)} + 6H_2O_{(\square)}$,

ΔH = positive. At equilibrium which factor will not effect the concentration of NH_3 is.

(1) Change in pressure

(2) Change in volume

(3) Addition of Catalyst

(4) Change in temperature

6) Consider the reactions given below; on the basis of these reactions find out correct relation between x and y.
$C_{(g)} + 4H_{(g)} \rightarrow CH_{4(g)}$; ΔH = –x kJ mol^{-1}
$C_{(graphite)} + 2H_{2(g)} \rightarrow CH_{4(g)}$; ΔH = –y kJ mol^{-1}

(1) x = y

(2) x = 2y

(3) x > y

(4) x < y

7)

Consider the statements given below :
(A) Addition of catalyst during chemical reaction alters activation energy.
(B) If two different rate of I^{st} order reaction at different time is given then half life period can be calculated.
(C) Decomposition of phosphine (PH_3) on tungston at low pressure is I^{st} order reaction.
(D) Rate of reaction increases with decreases in concentration of reactant.
(E) $t_{1/2}$ of zero order reaction is independent of temperature.
Choose the number of correct statements from above statements.

(1) 2

(2) 3

(3) 4

(4) 5

8)

When acidified solution of ferrous ammonium sulphate (FAS) is treated with $KMnO_4$ solution in acidic medium then incorrect statement is :

(1) MnO_4^- is reduced to Mn^{2+} but Fe^{2+} will be oxidised to Fe^{3+}.

(2) 100 ml $\frac{N}{10}$ $KMnO_4$ solution requires 50 ml $\frac{N}{5}$ FAS solution for its reduction.

(3) NH_4^+ will be oxidised to NO_2 but MnO_4^- will be reduced to MnO_4^{2-}.

(4) $KMnO_4$ is an oxidising agent but FAS is reducing agent.

9) For the hypothetical reaction
$A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}$
$\Delta_rG°$ and $\Delta_rS°$ are 20 kJ/mol and –20 JK^{-1} mol^{-1} respectively at 200 K.
If Δ_rC_p is 20 JK^{-1} mol^{-1} then $\Delta_rH°$ at 400 K is :

(1) 20 kJ/mol

(2) 7.98 kJ/mol

(3) 28 kJ/mol

(4) 24 kJ/mol

10)

The half cell reactions for corrosion is given below :
$2H^+ + \frac{1}{2}O_2 + 2e^- \rightarrow H_2O_{(\square)}$; $E^0 = 1.23$ V
$Fe^{2+} + 2e^- \rightarrow Fe$; $E^0 = -0.44$ V
What will be value of ΔG^0(KJ) for overall reaction?
$Fe + \frac{1}{2}O_2 + 2H^+ \rightarrow Fe^{+2} + H_2O$ (□)

(1) –432

(2) 212

(3) –167

(4) –322

11)

The degree of dissociation of water at 25°C is found to be 1.8×10^{-9} what will be ionization constant of water at 25°C.

(1) 10^{-14}

(2) 1.8×10^{-14}

(3) 1.8×10^{-16}

(4) 4.26×10^{-17}

12) Vapour pressure of pure liquids A and B are 450 and 700 mm of Hg respectively at 350 K, then which is correct value of composition in liquid or vapour phase, if vapour pressure of solution is 600 mm of Hg

(A) $X_A = 0.4$ (B) $Y_A = 0.4$

(C) $X_B = 0.6$ (D) $Y_B = 0.7$

Choose the correct answer from the options given below :

(1) B only

(2) D only

(3) A, C and D only

(4) A, B and C only

13) Molality of 'a' gram solution is 'b'. If the mass of solute is 'c' gram then molar mass of solute in $gmol^{-1}$ is :-

(1) $\frac{c}{a} \times \frac{1000}{(b-a)}$

(2) $\frac{b}{a} \times \frac{1000}{(a-b)}$

(3) $\frac{(a-c)}{1000} \times \frac{b}{c}$

(4) $\frac{c}{b} \times \frac{1000}{(a-c)}$

14) From the following E° values of half cells,

(i) $A + e^- \rightarrow A^-$; E° = –0.24 V

(ii) $B^- + e^- \rightarrow B^{2-}$; E° = +1.25 V

(iii) $C^- + 2e^- \rightarrow C^{3-}$; E° = –1.25 V

(iv) $D + 2e^- \rightarrow D^{2-}$; E° = +0.68 V

which combination of two half cells would result in minimum cell potential ?

(1) (ii) and (iii)

(2) (ii) and (iv)

(3) (i) and (iii)

(4) (i) and (iv)

15) **Statement-I :** For the reaction $2O_3 \rightarrow 3O_2$, the rate law is : $r = K[O_3]^2[O_2]^{-1}$ mechanism :

Step-1 $O_3 \rightleftharpoons O_2 + O$ (fast)

Step-2 $O_3 + O \rightarrow 2O_2$ (slow)

Statement-II : The reaction has –ve order with respect to O_2.

(1) Both statement-I and statement-II are true.

(2) Both statement-I and statement-II are false.

(3) If statement-I is true but statement-II is false.

(4) If statement-I is false but statement-II is true.

16)

Match **List-I** with **List-II** :

	List-I (Test/reagent)		**List-II (Radical identified)**
A.	Lake Test	I.	NO_3^-
B.	Nessler's Reagent	II.	Fe^{3+}
C.	Potassium sulphocyanide (thiocyanate)	III.	Al^{3+}
D.	Brown Ring Test	IV.	NH_4^+

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-III, D-I

(2) A-II, B-IV, C-III, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-III, B-IV, C-II, D-I

17) Select the incorrect order of given properties :-

(1) B > T□ > Ga > A□ > In : Ionisation energy

(2) B > A□ < Ga < In < T□ : Electronegativity

(3) $B^{+3} > Al^{+3} > Ga^{+3}$: Lewis acidic strength

(4) B < A□ < Ga < In < T□ : Atomic size

18) Identify correct match .

	Atomic No.	**Name**
(1)	108	Bohrium
(2)	112	Rontgenium
(3)	106	Seaborgium
(4)	116	Moscovium

(1) 1

(2) 2

(3) 3

(4) 4

19) Pick the incorrect statement for crystalline allotropes of carbon.

(1) Largest bond length is present in diamond.

(2) Density of diamond is maximum.

(3) In fullerene (C_{60}) number of hexagon ring are greater than number of pentagon ring.

(4) All the three allotropes have dangling bonds.

20) Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : PH_3 has lower boiling point than NH_3.
Reason (R) : In liquid state NH_3 molecules are associated through vander waal's forces, but PH_3 molecules are associated through hydrogen bonding.
In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

(2) (A) is not correct but (R) is correct

(3) Both (A) and (R) are correct but (R) is the correct explanation of (A)

(4) (A) is correct but (R) is not correct

21) Which of the following option is correct ?

(1) Ce^{+4} solution are widely used as oxidizing agent in volumetric analysis

(2) Ce^{+4} is diamagnetic ion while Yb^{+2} is paramagnetic ion

(3) All the lanthanides are much more reactive than aluminium

(4) Both (2) and (3) are correct

22) Match **List-I** with **List-II** :

List-I (Element)		**List-II (Most Common oxidation state/s)**	
A.	Fe	I.	+2, +7
B.	V	II.	+3, +2
C.	Mn	III.	+4
D.	Th	IV.	+5

Choose the **correct** answer from the options given below :
(1) A-IV, B-II, C-I, D-III

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-IV, C-II, D-III

(4) A-II, B-IV, C-I, D-III

23) Choose the correct set representing the given statements as true (T) or false (F).
(I) Coordination isomerism arises from the interchange of ligands between cationic and anionic entities present in a complex.
(II) Thiocyanate ligand is an ambidentate ligand.
(III) Ionisation isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then becomes the counter ion.
(IV) $[Co(NH_3)_6][Co(CN)_6]$ do not exhibit coordination isomerism

(1) FTTT

(2) TTTT

(3) TTTF

(4) FTFT

24) Identify the incorrect statement:-

(1) The magnetic behaviour of metal ions having upto three electrons in the d orbitals in their coordination entities is independent on the nature of ligand.

(2) Complex ions having d^4, d^5, d^6 & d^7 configuration with presence of SFL are low spin complexes.

(3) Magnetic moment of coordination compounds is independent on nature of ligands.

(4) $[Mn(CN)_6]^{-3}$ has magnetic moment of two unpaired electrons whereas $[MnCl_6]^{-3}$ has four unpaired electrons.

25) Correct statement is.
(A) Complex $[Pt(NH_3)_2Cl_2]$ show geometrical isomerism.
(B) Optical isomers are mirror images that can not be superimposed on one another. These are called enantiomers.
(C) Chlorophyll is a complex compound of magnesium.
(D) Ethylenediamine is a mono dentate ligand.

(1) (A), (B), (C), (D) are correct options.

(2) (A), (B), (C) are correct options.

(3) (A), (B), (D) are correct options.

(4) (A), (D) are correct options.

26)

Match the column

Column I (Solid)		**Column II (Examples)**	
a	Covalent	P	SiO_2
b	Molecular	Q	CaO
c	Ionic	R	CCl_4
d	Metallic	S	Zn

(1) (a) P (b) Q (c) R (d) S

(2) (a) R (b) P (c) Q (d) S

(3) (a) S (b) P (c) Q (d) R

(4) (a) P (b) R (c) Q (d) S

27) Which of the following statement are true and false ?
(a) In PCl_5 hybridisation is sp^3d and it has a trigonal pyramidal structure.
(b) The angle between the P-Cl bonds is 90°, which is same for all the P and Cl present in PCl_5
(c) The bond length of P-Cl in axial position is higher than in equitorial position
(d) PCl_5 have zero dipole moment

Choose the correct option.

(1) TFTT
(2) FTTT
(3) FFTT
(4) TFFT

28) Which of the following statement is not correct

(1) Inert gases have a different number of outermost electrons than their group number
(2) In the sixth period the orbitals being filled are 6s, 4f, 5d, 6p
(3) The second transition series contains the elements from Y to Cd.
(4) Inert gases are soluble in water due to keesom force.

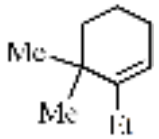
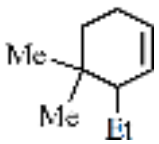
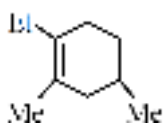
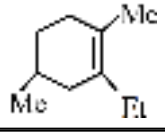
29) ClO_2 is an anhydride of -

(1) Chlorous acid
(2) Chloric acid
(3) Both 1,2
(4) Perchloric acid

30) Select correct order :-

(1) O > N : I.P. Order
(2) $N_2 > N_2^- > N_2^{\oplus}$: Stability
(3) $B_2 < C_2$: Stability
(4) N – N > P – P : Catenation property

31) Incorrect IUPAC name is :

	Compound	IUPAC name
(A)		1-Ethyl-6, 6-dimethyl cyclohexene
(B)		2-Ethyl-3, 3-dimethyl cyclohexene
(C)		1-Ethyl-2, 4-dimethyl cyclohexene
(D)		2-Ethyl-1, 4-dimethyl cyclohexene

(1) A

(2) B

(3) C

(4) D

32) **Statement-1 :** Ethanol gives iodoform test with iodine & alkali.
Statement-2 : All type of primary alcohol shows iodoform test.

(1) Both Statement-1 and Statement-2 are correct.

(2) Both Statement-1 and Statement-2 are incorrect.

(3) Statement-1 is correct and Statement-2 is incorrect.

(4) Statement-1 is incorrect and Statement-2 is correct.

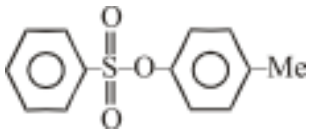
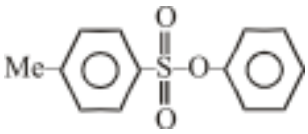
33) The incorrect statement about glucose is :

(1) Glucose and fructose are the monomer unit in sucrose.

(2) Glucose is an aldohexose.

(3) Glucose have five chiral carbon in cyclic form.

(4) Glucose is insoluble in water because of having aldehyde functional group.

34) Which type of isomerism is shown by :  & 

(1) Chain isomerism

(2) Position isomerism

(3) Metamerism

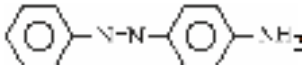
(4) Functional group isomerism

35)

The incorrect match is :

(1) Hinsberg reagent - $C_6H_5SO_2Cl$

(2) Stephen's reducing agent –$SnCl_2$ + HCl

(3) Aniline yellow – 

(4) Reagent which is used in clemmensen reduction – H_2/Pd + $BaSO_4$

36) Lassiagne solution when react with sodium nitropruside, the violet colour complex appear, it shows presence of :-

(1) Nitrogen

(2) Phosphorus

(3) Sulphur

(4) Chlorine

37)

Which of the following statement is incorrect for S_N2 reaction :-

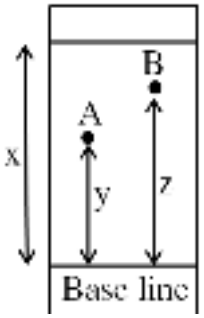
(1) S_N2 is one step process

(2) During S_N2 reaction inversion in configuration takes place

(3) Rate of reaction depend on concentration of nucleophile as well as substrate

(4) Reaction proceeds through formation of carbocation intermediate.

38) A chromatogram is shown below after TLC :



Here
x = Distance covered by solvent = 12 cm
y = Distance covered by A component = 6 cm
z = Distance covered by B component = 8 cm
Find R_f value of A component ?

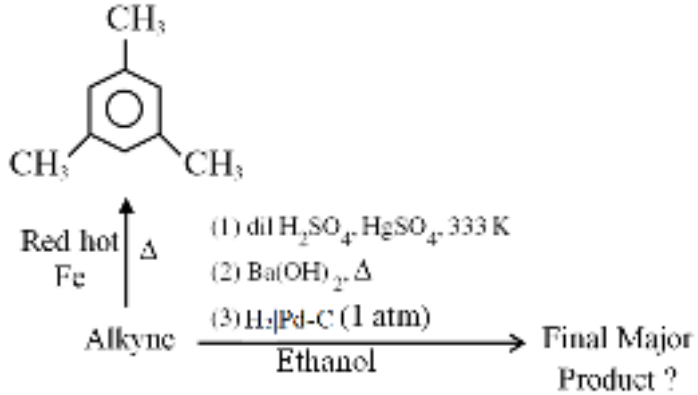
(1) 2

(2) 0.5

(3) 1.5

(4) 0.66

39)



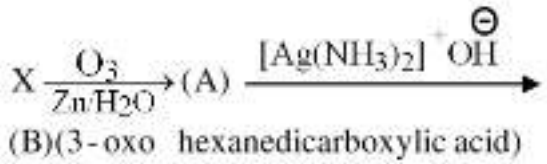
(1) $CH_3-\overset{\underset{O}{\|}}{C}-CH=\overset{\underset{CH_3}{|}}{C}-CH_3$

(2) $CH_3-\overset{\underset{OH}{|}}{CH}-CH_2-\overset{\underset{CH_3}{|}}{CH}-CH_3$

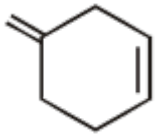
(3) $CH_3-\overset{\underset{O}{\|}}{C}-CH_2-\overset{\underset{CH_3}{|}}{CH}-CH_3$

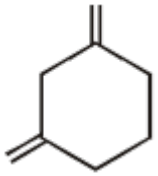
(4) $CH_3-\overset{\underset{OH}{|}}{CH}-CH=\overset{\underset{CH_3}{|}}{C}-CH_3$

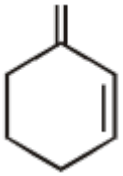
40) An unsaturated hydrocarbon X absorbs two hydrogen molecules on catalytic hydrogenation, and also gives following reaction :

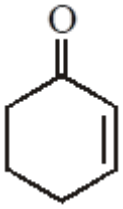


X will be :-

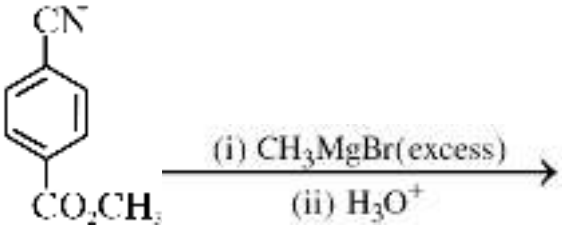
(1) 

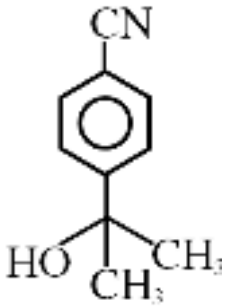
(2) 

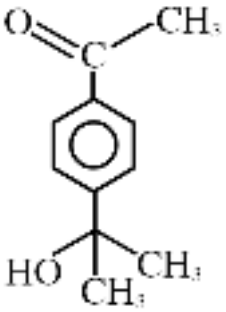
(3) 

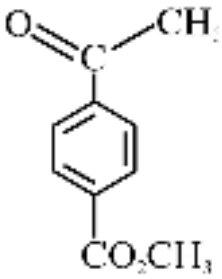
(4) 

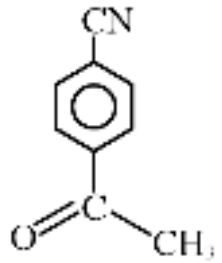
41) Major product of the following reaction is



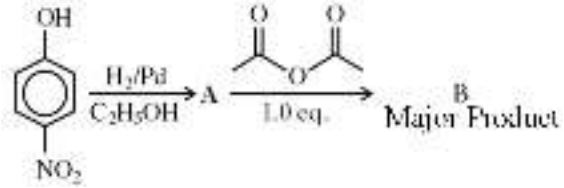
(1) 

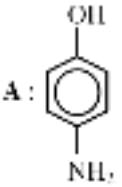
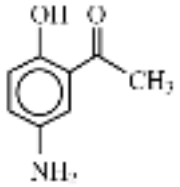
(2) 

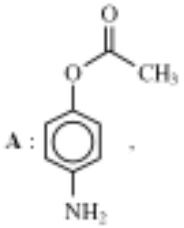
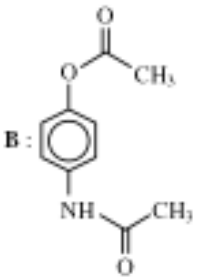
(3) 

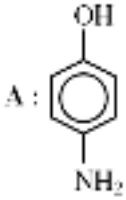
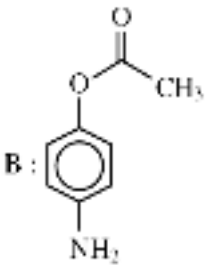
(4) 

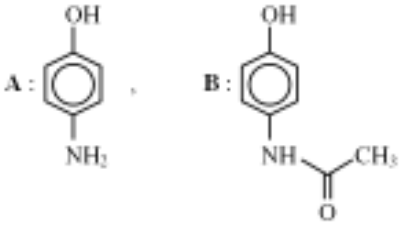
42) The correct structures of **A** and **B** formed in the following reactions are :



(1) **A :**  , **B :** 

(2) **A :**  , **B :** 

(3) **A :**  , **B :** 

(4) 

43) Which of the following has highest boiling point ?

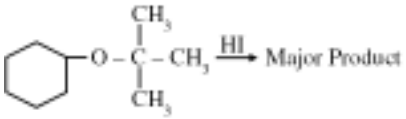
(1) $CH_3CH_2CH_2CH_3$

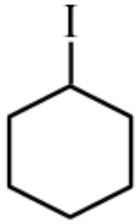
(2) $CH_3CH_2CH_2CH_2$–OH

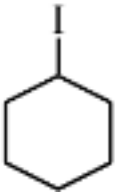
(3) $CH_3CH_2CH_2CHO$

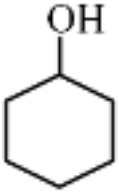
(4) H_5C_2 – O – C_2H_5

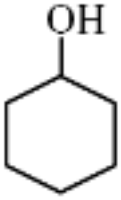
44) Major product formed in the following reaction is a mixture of:



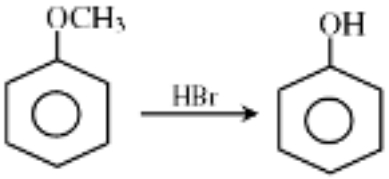
(1)  and $(CH_3)_3Cl$

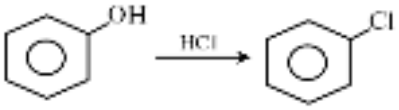
(2)  and $(CH_3)_3COH$

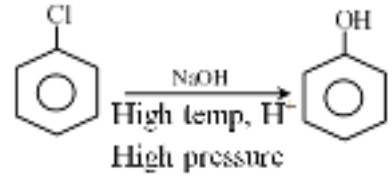
(3)  and $(CH_3)_3COH$

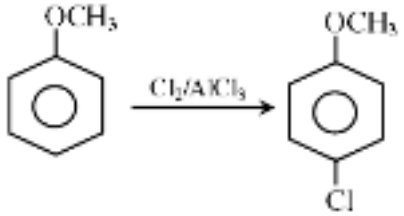
(4)  and $CH_3-\overset{CH_3}{\underset{CH_3}{C}}-I$

45) Which one of the following reactions is NOT possible?

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Given below are two statements:
Statement-I: Chondrichthyes fishes have cartilaginous endoskeleton, mouth located ventrally, Notochord persistent throughout the life, gill slits without operculum. Predaceous animal with powerful jaw and air bladder absent.
Statement-II: Osteichthyes fishes found in both fresh water & marine habitat, mouth is terminal gills covered by operculum on each side fertilization usually external
In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both statement-I and statement-II are incorrect
(2) Statement-I is incorrect but statement-II is correct
(3) Statement-I is correct but statement-II is incorrect
(4) Both statement-I and statement-II are correct

2)

Study the following table & fill in the blanks.

Organism	**Excretory organ**
Nereis	(i)
(ii)	Malpighian tubule
Balanoglossus	(iii)
(iv)	Kidney

(1) (i) Nephridia, (ii) Cockroach,
(iii) Proboscis gland, (iv) Rana

(2) (i) Protonephridia, (ii) Cockroach,
(iii) Proboscis gland, (iv) Rana

(3) (i) Nephridia, (ii) Pila,
(iii) Gills, (iv) Petromyzon

(4) (i) Nephridia, (ii) Star fish,
(iii) Gills, (iv) Rana

3) Match the column-I with column-II on the basis of phylum and example ?

Column-I (Phylum)		**Column-II (Example)**	
A	Porifera	i	Pleurobrachia
B	Cnidaria	ii	Taenia
C	Ctenophora	iii	Spongilla
D	Platyhelminthes	iv	Aurelia

(1) A-iii, B-iv, C-i, D-ii

(2) A-i, B-iii, C-ii, D-iv

(3) A-iv, B-ii, C-i, D-iii

(4) A-ii, B-i, C-iv, D-iii

4) **Assertion (A):** Most distinctive feature of echinoderm is presence of water canal system.
Reason (R) : Water canal system help in locomotion, capture & transport of food & respiration.

(1) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.

(2) **Assertion** and **Reason** both are incorrect.

(3) **Assertion** is incorrect but **Reason** is correct.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

5) In ____a_____ Notochord present only in larval tail. While in ____b____ it extends from head to tail throughout life

(1) a - Cephalo chordata, b- Urochordata

(2) a-Urochordata, b-Cephalochordata

(3) a - Protochordata, b- Tunicata

(4) a - Tunicata, b-Urochordata

6) Which of the statement is incorrect regarding reptiles

(1) They have external ear opening

(2) They are Oviparous

(3) This class have crawling mode of locomotion

(4) They are poikilotherms.

7) A researcher is studying the epithelium lining the urinary bladder. She notices that despite the bladder stretching and relaxing frequently, the tissue does not leak urine between the cells.
Which junction type is most likely preventing leakage between epithelial cells in this tissue?

(1) Gap junctions
(2) Desmosomes
(3) Tight junctions
(4) Adherens junctions

8) In cockroach small non chitinous flexible structure that lies on the floor of preoral cavity. Identify the structure :

(1) Labrum
(2) Labium
(3) Hypopharynx
(4) Mandibles

9) Homo sapiens and Rana tigrina have similar

(1) Number of chambers in heart
(2) Number of cranial nerves
(3) Main excretory product
(4) Special venous connection present between kidney and lower parts of the body

10) **Assertion:** Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.
Reason: Atherosclerosis affects the vessels that supply blood to the heart muscles, it is caused by deposits of calcium, fat , cholesterol and fibrous tissues which makes the lumen of arteries narrower.

(1) 'A' and 'R' both correct and 'R' is the correct explanation of 'A'
(2) 'A' and 'R' both correct but 'R' is not the correct explanation of 'A'
(3) 'A' is correct but 'R' is incorrect
(4) 'A' is incorrect but 'R' is correct

11) If a person has blood group B negative, which of the following blood groups would be the safest for transfusion?

(1) B positive and O negative
(2) B negative and O negative
(3) AB negative and A negative
(4) O positive and B positive

12) Respiratory rhythm is controlled by medullary rhythm centre. Pneumotaxic centre in the pons reduces duration of inspiration. Chemo sensitive area responds primarily to CO_2 and H^+ not O_2. A slight rise in PCO_2 strongly stimulates breathing.
If Pneumotaxic centre becomes hyperactive which change will occurs ?

(A) Inspiration becomes shorter
(B) Respiratory rate increase
(C) Tidal volume increases
(D) Alveolae ventilation decreases

(1) A, B
(2) B, C
(3) A, D
(4) C, D

13) Choose the incorrect statement:-

(1) Every 100 ml impure blood delivers 4ml of CO_2 to the tissue
(2) 20–25% of CO_2 is transported as carbamino haemoglobin
(3) Binding of gases is related to partial pressure
(4) RBCs contain enzyme carbonic anhydrase.

14) A person was exposed to conditions which led to prolonged sweating and significant loss of body fluid. This led to enhanced sodium and water reabsorption in later parts of nephron, mediated by activation of a specific adrenal hormone that is

(1) Cortisol
(2) Adrenalin
(3) ANF
(4) Aldosterone

15) Which of the following is incorrect?

(1) Tubular secretion helps in maintaining ionic balance.
(2) PCT reabsorbed 70-80% glucose actively.
(3) Glucose, Amino acid & Na^+ are reabsorbed actively.
(4) GFR in healthy person is 125 ml/min.

16) Read the paragraph:
"The axonal membrane at rest is more permeable to K^+ ions and impermeable to negatively chargec proteins ... "
Which of the following statements is correct?

(1) Axoplasm has more Na+ than outside fluid.
(2) Outside fluid has more Na+ and less K^+.
(3) Resting potential is positive on inside.
(4) Na^+– K^+ pump sends 2 Na^+ out and 3 K^+ in.

17) **Statement-I :-** The layer of cells which covers the cerebral hemisphere is called cerebral cortex and is thrown into prominent folds.
Statement-II :- Fibres of the tracts are covered with the myelin sheath, which constitute the inner part of cerebral hemisphere.

(1) Both statement-I and statement-II are incorrect

(2) Both statement-I and statement-II are correct

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

18) (a) Pineal secretes a hormone called melanin.
(b) Deficiency of iodine causes goitre.
(c) Deficiency of ADH causes type-I diabetes
(d) PTH is a hypercalcemic hormone
Read the above statements and choose the correct option:

(1) a & b

(2) c & d

(3) b & d

(4) a & d

19) Hyper secretion of which hormone causes myasthenia gravis ?

(1) Thyroxine

(2) Thymosine

(3) Adrenalin

(4) Cortisol

20) Read the following statement carefully -
(A) Rapid transmission of information
(B) Slow, long lasting response
(C) Non-nutrient chemicals
(D) Act as intracellular messenger
(E) Released into the blood and lymph
(F) Produced in trace amount
(G) Have specific target cells
(H) Regulate growth, metabolism and reproduction.
How many statement are incorrect for hormones.

(1) One

(2) Two

(3) Three

(4) Four

21) Which one of the following statement is incorrect ?

(1) Heart muscles are striated and involuntary

(2) The muscles of hand and legs are striated and voluntary

(3) The muscles located in the inner wall of alimentary canal are striated and involuntary

(4) Muscles located in the reproductive tract are unstriated and involuntary.

22) Smooth muscle fiber are found in the walls of internal organs such as _____, stomach and

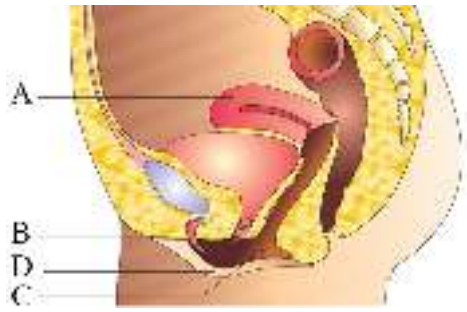
intestine.

(1) Bones
(2) Skin
(3) Blood vessels
(4) Brain

23) Which of the following bones are not included in the axial skeleton ?

(1) Cranial bones
(2) Ribs
(3) Girdle bones
(4) Facial bones

24) Identify A, B, C and D labelling from the options given below :



(1) A - Uterus; B - Clitoris; C - Labia majora;
D - Labia minora

(2) A - Uterus; B - Clitoris; C - Vagina;
D - Labia majora

(3) A - Urinary bladder; B - Clitoris; C - Vagina;
D - Labia minora

(4) A - Urinary bladder; B - Vagina; C - Clitoris;
D - Uterus

25) Read the following paragraph:
The primary sex organs – the testis in the males and the ovaries in the females which produces gametes. (i.e, sperms and ovum) respectively, by the process called gametogenesis. There are remarkable differences between the reproductive events in the male and in the female, for example, sperm formation continues even in old men, but formation of ovum ceases in women around the age of fifty years.
Given below are two statements about this:
Statement I: The spermatogonia present on the inside wall of seminiferous tubules multiply by mitotic division and increase in numbers. Each spermatogonium is haploid and contains 23 chromosomes.
Statement II: The secondary follicle soon transforms into a tertiary follicle which is characterised by a fluid filled cavity called antrum. The theca layer is organised into an inner theca interna and an outer theca externa.
In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both the statements are correct
(2) Both the statements are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

26) Match column I with column II and select the correct option using the codes given below :-

	Column-I		**Column-II**
(a)	Corpus luteum	(i)	Germinal layers
(b)	Clitoris	(ii)	Sperm
(c)	Nebenkern sheath	(iii)	Female external genitalia
(d)	Inner cell mass	(iv)	Endocrine structure

	a	**b**	**c**	**d**
(1)	iv	iii	ii	i
(2)	i	iii	ii	iv
(3)	i	ii	iii	iv
(4)	iv	ii	iii	i

(1) 1

(2) 2

(3) 3

(4) 4

27) Which of the following hormone leads to expulsion of the baby out of the uterus through birth canal ?

(1) Prolactin

(2) Progesterone

(3) Estrogen

(4) Oxytocin

28) **Assertion (A):** Both bone marrow and thymus provide micro-environments for the development and maturation of B-lymphocytes.
Reason (R): The thymus is quite large at the time of birth but keeps reducing in size with age and by the time puberty is attained it reduces to very small size.
In the light of above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

29) Read the following statements:
a - Active immuity is slow and takes time to give its full effective response

b - The injection of Anti venom is an example of passive immunisation
c - 'The humoral immune response' is produced by T-lymphocytes
d - The health is simply mean absence of diseases" or "Physical fitenss"
Select the option which includes all the correct statements only:

(1) a, b and c

(2) b, c and d

(3) a and b only

(4) b and c only

30) Group of chemicals which interact with receptors present principally in the brain. Generally taken by inhalation and oral ingestion, these are Known for their effects on cardiovascular system of the body. These chemicals are:-

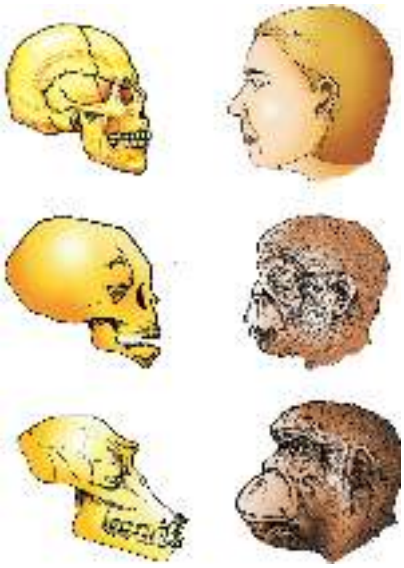
(1) Morphine

(2) Cannabinoids

(3) Cocaine

(4) Barbiturates

31)



What conclusion can be drawn from the diagram given above?

(1) The similarity between the baby chimpanzee skull and the adult human skull supports the ideal of a common evolutionary ancestor for humans and chimpanzees.

(2) Humans evolved directly from modern monkey.

(3) Humans and chimpanzees do not share any evolutionary relationship.

(4) Modern humans evolved directly from baby chimpanzees.

32) According to Oparin and Haldane, the primitive atmosphere of early earth was predominantly

(1) Oxidising with free oxygen

(2) Reducing with methane and ammonia

(3) Neutral with nitrogen only

(4) Rich in ozone

33) Match the list-I with list II :-

	List-I		List-II
(a)	Chlorophyll	(i)	Chromatin
(b)	70s Ribosome	(ii)	Cytoplasm
(c)	Axoneme	(iii)	Stroma
(d)	Kinetochore	(iv)	Cilia
		(v)	Thylakoids
		(vi)	Chromosome
		(vii)	Centriole

	a	b	c	d
(1)	(iii)	(ii)	(iv)	(vi)
(2)	(v)	(iii)	(iv)	(i)
(3)	(v)	(iii)	(iv)	(vi)
(4)	(iii)	(ii)	(vii)	(vi)

(1) 1

(2) 2

(3) 3

(4) 4

34) Select the correct statement -
(A) Dissolution of synaptonemal complex occur during diplotene.
(B) Prophase is marked by the initiation of condensation of chromosomal material.
(C) Chromosome cluster at opposite spindle poles and their identity is lost as discrete element during metaphase of mitosis.
(D) Bivalent or tetrad become clearly visible during pachytene.
(E) Homologous chromosomes separate, while sister chromatids remain associated at their centromere occur during metaphase-I of meiosis.
Choose the correct answer from the option given below :

(1) A, B & E only

(2) A, B & D only

(3) B, C & E only

(4) C, D & E only

35) Match List-I with List-II.

List-I (Events)		List-II (Prophase-I)	
(A)	Terminalisation of chiasmata take place	(I)	Zygotene

(B)	Crossing over take place between non-sister chromatids of homologous chromosome.	(II)	Pachytene
(C)	X-hasped structure called chiasmata become visible.	(III)	Diakinesis
(D)	Chromosomes becomes gradually visible under light microscope and compaction continues	(IV)	Diplotene

Choose the correct answer from the options given below:

(1) A-III, B-II, C-IV, D-I

(2) A-IV, B-II, C-I, D-III

(3) A-II, B-III, C-I, D-IV

(4) A-I, B-II, C-III, D-IV

36) **Assertion :** In some organisms, karyokinesis is not followed by cytokinesis as a result of which multinucleated condition on arise leading to the formation of a syncytium (e.g. liquid endosperm in coconut).
Reason : Cytokinesis in plants is by cell-plate method.

(1) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

37) How many of the following statements are correct?
(a) All cells arise from pre-existing cells.
(b) Ribosomes and centriole are non-membrane bound organelles found in cells both eukaryotic as well as in prokaryotic.
(c) Organelle within organelle-mitochondria and ribosome.
(d) Plasma membrane of bacteria is selectively permeable

(1) Four

(2) One

(3) Three

(4) Two

38) Read both statement and select the correct option :
Statement - I :- During different stages of cell division, cell show structured chromosomes in place of the nucleus.
Statement - II :- The telocentric chromosome has a terminal centromere.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

39) 

In given figure of longitudinal section of mitochondria, 'A' represents?

(1) Outer membrane

(2) Matrix

(3) Inter membrane space

(4) Crista

40) Which of the following statement is incorrect regarding restriction endonucleases.

(1) Restriction enzyme belongs to larger class of enzymes called nucleases.

(2) Each restriction endonuclease functions by inspecting the length of a DNA segment.

(3) Restriction enzymes cut the strand of DNA only in the centre of palindromic site.

(4) Restriction endonucleases are used in genetic engineering to form recombinant molecule of DNA.

41) Match the following :-

Column-I		**Column-II**	
(A)	First artificially synthesized hormone	(i)	Tobacco
(B)	First transgenic plant	(ii)	Adenosine deaminase deficiency
(C)	First clinical gene therepy	(iii)	Insulin
(D)	RNAi	(iv)	Cellular defense

(1) A-iii, B-i, C-iv, D-ii

(2) A-iii, B-i, C-ii, D-iv

(3) A-ii, B-i, C-iv, D-iii

(4) A-ii, B-iii, C-i, D-iv

42) Which of the following is wrongly matched in the given table ?

(i)	Golden rice	Transgenic rice	Vitamin A enriched
(ii)	Flavr savr	Transgenic tobacco	Delayed ripening
(iii)	Bt-cotton	Transgenic cotton	Bacterial and viral resistance
(iv)	Rosie	Transgenic cow	Milk contained human protein alpha antitrypsin

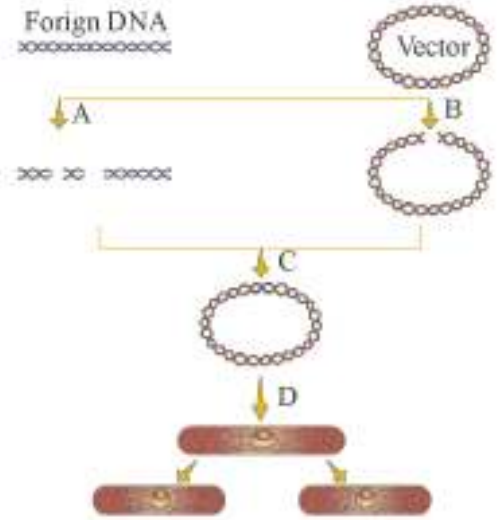
(1) i and ii

(2) ii, iii and iv

(3) ii and iii

(4) Only iii

43) The diagram below refers to recombinant DNA technology. Identify A to D:



	A	B	C	D
(1)	EcoRI	Hind II	DNA ligase	Transformation
(2)	EcoRI	EcoRI	DNA ligase	Transformation
(3)	Hind II	EcoRI	DNA ligase	Transformation
(4)	EcoRI	EcoRI	Transformation	DNA ligas

(1) 1

(2) 2

(3) 3

(4) 4

44) If recombinant DNA carrying antibiotic resistance gene (e.g. ampicillin) is transferred into *E.coli* cell, the host cell is transformed into ampicillin resistant cells. The ampicillin resistant gene in this case is called a :

(1) Vectors

(2) Plasmid

(3) Selectable marker

(4) Cloning sites

45) **Statement-I:** Lipases are used in detergent formulations and are helpful in removing oily stairs from laundry.
Statement-II: Antibiotic are chemical substance which are produced by some microbes and can kill or retarded the growth of microbes.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

46) Which option is related with only monera and not related with other kingdom of R.H. Whittaker?

(1) Decomposer nature

(2) Presence of cell wall and cell membrane

(3) Well developed tissue system

(4) Nitrogen fixation ability

47) Match the column-I, column-II and column-III and select the correct option from options given below :

Column-I (Types of spores)		**Column-II (Fungus)**		**Column-III (Fungi class)**	
(A)	Zoospores	(P)	*Penicillium*	(i)	Basidiomycetes
(B)	Conidia	(Q)	*Albugo*	(ii)	Deuteromycetes
(C)	Ascospore	(R)	*Agaricus*	(iii)	Ascomycetes
(D)	Basidiospore	(S)	*Trichoderma*	(iv)	Phycomycetes

(1) A-S-ii, B-P-iii, C-Q-iv, D-R-i

(2) A-Q-iv, B-P-iii, C-S-ii, D-R-i

(3) A-Q-iv, B-S-ii, C-P-i, D-R-iii

(4) A-Q-iv, B-S-ii, C-P-iii, D-R-i

48) Consider the following statements :-
(a) Presence of phycobilins
(b) No motile stage
(c) Prokaryotic cell structure
(d) Presence of heterocyst
(e) Formation of hormogonia and akinetes
How many of the above statements are only true for Cyanobacteria and are not true for red algae ?

(1) Three

(2) Four

(3) Five

(4) Two

49) **Statement-I** : Gemmae are green, unicellular, asexual buds, which develop in small receptacles called gemma cups located on the thalli.
Statement-II : In liverworts, spores germinate to form free-living sporophytes

(1) Both **Statement I** and **Statement II** are incorrect.

(2) Only **Statement I** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) Only **Statement II** is correct.

50) The given table shows classes of algae and their main characteristics. Some characteristics in this table mentioned as A, B, C and D. Observe the table and choose the option in which A, B, C and

D are correctly identified :-

Classes	Chlorophyceae	Phaeophyceae	Rhodophyceae
Major Pigments	Chl a, b	Chl a, c	**A**
Stored Food	Starch	**B**	Floridean starch
Cell Wall	**C**	Cellulose	Cellulose, pectin, polysulphate ester
Flagellar Number	2-8	2	**D**

	A	B	C	D
(1)	Chl a, e	Fat & oil	Peptidoglycan	Many
(2)	Chl a, d	Mannitol, Laminarin	Cellulose	Absent
(3)	Chl a, d	Paramylum	Lipoprotein	One
(4)	Chl a, c	Glycogen	Cellulose	Absent

(1) 1
(2) 2
(3) 3
(4) 4

51) Algae in which sexual reproduction is of anisogamous type :-

(1) *Volvox*
(2) *Eudorina*
(3) *Spirogyra*
(4) *Ulothrix*

52) Which of the following statement is **true** ?

(1) In pinnately compound leaf, leaflets are attached at common point.
(2) A bud is absent in the axil of petiole in compound leaves.
(3) Phyllotaxy is the pattern of arrangement of leaves on the stem or branch
(4) In pinnately compound leaves are found in silk cotton.

53)

Consider the following four statements (a-d) and select the option which includes all the correct ones :
(a) In some leguminous plants, the leafbase may become swollen, which is known as pulvinus.

(b) Long thin flexible petioles allow leaf blades to flutter in wind there by cooling the leaf and bringing fresh air to leaf surface.
(c) A single leaf arises at each node in mustard and chinarose.

(d) Leaves originate from shoot apical meristem and arranged in basipetal manner.

(1) a, b and c

(2) b, c and d

(3) a and b only

(4) c and d only

54) Identify the correct match from the column-I, II and III

Column-I	**Column-II**	**Column-III**
(1) Zygomorphic flower	(a) Irregular	(i) Cucumber, sunflower
(2) Perigynous flower	(b) Inferior ovary	(ii) *Canna*
(3) Epigynous flower	(c) Bilateral symmetry	(iii) Rose, plum
(4) Asymmetric flower	(d) Half-inferior ovary	(iv) Pea, bean

(1) 1-c-iv, 2-a-iii, 3-b-i, 4-d-ii

(2) 1-c-iv, 2-d-iii, 3-b-i, 4-a-ii

(3) 1-d-iii, 2-b-iv, 3-c-i, 4-a-ii

(4) 1-c-iv, 2-a-iii, 3-b-ii, 4-d-i

55) Identify the correct statement :

(1) Family has a group of related order

(2) Genus comprises a group of related species which has more characters in common comparision to species of other genera

(3) Classification is a single step process

(4) *Petunia* and *Datura* are placed in the family Anacardiaceae

56) Find the incorrect from the following :-

(1) Sporophyte of moss is more elaborate than that in liverworts

(2) Spores of bryophytes germinate to form free living sporophytes

(3) Moss plants consist of upright, slender axes bearing spirally arranged leaves

(4) After meiosis spores are produced within the capsule of bryophytes

57) Which of the following statement is **not** correct about monocot root?

(1) Pith is large & well developed

(2) Secondary growth does not occur

(3) Xylem bundles are two to six in numbers

(4) It has epidermis, cortex, endodermis, pericycle, vascular bundles.

58) How many in the list given below are the part of ground tissue system ?
Epidermis, Root hair, Xylem, Phloem, Cortex, Pericycle, Pith

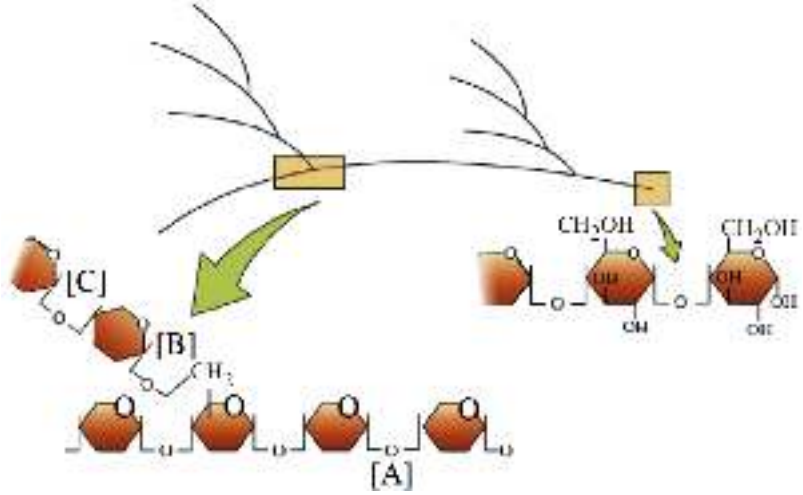
(1) One

(2) Two

(3) Three

(4) Four

59) In the given below diagrammatic representation of a portion of glycogen, glycosidic bonds



represented by A, B and C are :-

	A	B	C
(1)	α-1'-4"	α-1'-4"	β-1'-4"
(2)	β-1'-6"	β-1'-4"	β-1'-4"
(3)	α-1'-6"	α-1'-4"	α-1'-4"
(4)	α-1'-4"	α-1'-6"	α-1'-4"

(1) 1

(2) 2

(3) 3

(4) 4

60) Match the following:

Column-I		Column-II	
(A)	Pigment	(i)	Abrin
(B)	Lectins	(ii)	Anthocyanin
(C)	Drugs	(iii)	Vinblastin
(D)	Toxins	(iv)	Concanavalin A

(1) A–(iii), B–(i), C–(ii), D–(iv)

(2) A–(iv), B–(i), C–(ii), D–(iii)

(3) A–(iii), B–(i), C–(iv), D–(ii)

(4) A–(ii), B–(iv), C–(iii), D–(i)

61) Which of the following statements is **not** correct ?

(1) Carbonic anhydrase enzyme accelerate the rate of reaction $CO_2 + H_2O \rightarrow H_2CO_3$ by 10 million times.

(2) Functional aspect of enzymes is related to the tertiary structure of proteins

(3) Enzymes are composed of one or several polypeptide chains

(4) Low temperature does not cause the denaturation but preserves the enzyme in a permanent inactive state

62) If we remove accessory pigments from plant then what will be the effect ?

(1) The efficiency of photosynthesis will increase because all the light energy become available to chlorophyll 'a'

(2) Range of wavelengths, which are utilised in photosynthesis, become narrower

(3) Due to less amount of light energy the possibility of photo-oxidation of chlorophyll 'a' will decrease

(4) The photolysis of water and formation of ATP will not occur

63) **Statement-I :** The breaking of C-C bond of complex compound through oxidation within the cells, leading to release of considerable amount of energy is called photosynthesis.
Statement-II : During oxidation within a cell, all the energy contained in respiratory substrates is released in a single step.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

64) Which statement is correct for citric acid cycle?

(1) The cycle starts with condensation of acetyl group with OAA and water

(2) There are three points in the cycle where FAD^+ is reduced to $FADH_2$

(3) There is one point in the cycle where NAD^+ is reduced to $NADH + H^+$

(4) During conversion of α-ketoglutaric acid to succinyl CoA a molecule of GTP is synthesised

65) (A) GA – Extend market period of fruits (B) Ethylene – Increase surface area of absorption
(C) CK – Apical hook formation
(D) GA – Internodal elongation in deep water rice plant
How many matches are incorrect ?

(1) Four

(2) Three

(3) Two

(4) One

66) Which one of the following options given correct categorisation of three hormones.

	A = Auxin	B = GA	C = ABA
(1)	Human urine	Foolish seeding disease	Prevention of Abscisision
(2)	Rooting hormone	Bolting hormone	Stress hormone
(3)	Induce flowering	Inhibition of α-amylase	Flowering in LDP
(4)	Yabuta	F.W. went	Addicott et al

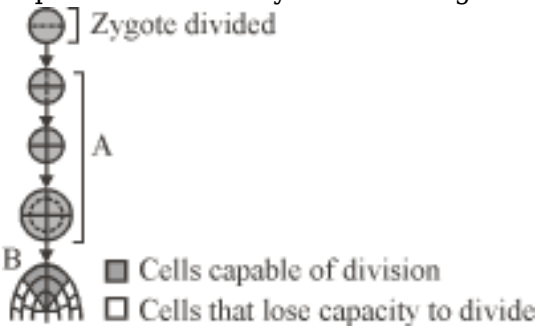
(1) 1

(2) 2

(3) 3

(4) 4

67) The given figure shows development of an embryo that undergoes two phases A and B. Select the correct option regarding it.



	A	**B**
(1)	Geometric phase	Arithmetic phase
(2)	Arithmetic phase	Geometric phase
(3)	Arithmetic phase	Exponential phase
(4)	Stationary phase	Stationary phase

(1) 1

(2) 2

(3) 3

(4) 4

68) In flowering plants male gametes are formed by:-

(1) Generative cell

(2) Vegetative cell

(3) Synergid cell

(4) Tube cell

69) In angiosperms following events occur during microgametogenesis :-
(a) Mitotic division in microspore.

(b) Meiotic division in miorospore mother cell.
(c) Formation of pollen grain.
(d) Formation of microspore.
(e) Formation of mature male gametophyte.
The correct sequence of event is :-

(1) b, a, d, e, c
(2) c, b, a, e, d
(3) b, d, a, c, e
(4) d, e, b, a, c

70) Which one of the following statement is not correct?

(1) Pollen grains are released from anthers at 2-celled stage
(2) Primary sporogenous cell directly behaves as the megaspore mother cell
(3) Megaspore divides twice to form eight nucleated embryosac
(4) Egg and synergids always lie near the micropylar end of ovule

71) In case of polygenic quantitative inheritance, if a trait is controlled by three genes, then in F_2 generation what will be the number of dominant alleles in the organism which is obtained in maximum number ?

(1) 6
(2) 4
(3) 3
(4) 2

72) Point mutation may occur due to

(1) Gain of a segment of DNA
(2) Deletion of segment of DNA
(3) Alteration in complete DNA sequence
(4) Change in a single base pair of DNA

73) **Assertion :** If single gene exhibit multiple phenotypic expression then it is an example or pleiotrophy.
Reason : In polygenic inheritance the effect of each allele is additive.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

74) If yellow body, white eyed *Drosophila* is crossed with wild brown body, red eye *Drosophila*, Then what would be the frequency of recombination in F_1 generation ?

(1) 100%

(2) 1.3%

(3) 98.7%

(4) 0%

75) Given below are some organisms:-
I - Human male
II - Female hen
III - Male Drosophila
IV - Male grasshopper
V - Male birds
Suppose
Male heterogamety = A
Female heterogamety = B
Male homogamety = C
On the basis of sex determination methods, which of the following combinations is correct?

Option:-

	A	**B**	**C**
(1)	I,III,IV	II	V
(2)	II, IV	I	III
(3)	III, IV	II	I
(4)	II, IV	I	II

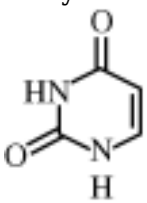
(1) 1

(2) 2

(3) 3

(4) 4

76) Identify the following molecule shown below with their nature and occurrence from the given table.



	Molecule	**Nature**	**Occurrence**
(1)	Uracil	Nucleotide	Found in RNA
(2)	Uridine	Nucleoside	Found in both RNA and DNA
(3)	Uracil	Nitrogen base	Found in RNA
(4)	Uridine	Nucleoside	Found in t-RNA

(1) 1

(2) 2

(3) 3

(4) 4

77)

Match List-I with List-II and select the correct answer :-

	List-I	List-II
a.	Ligase	Joins short segments of DNA together.
b.	DNA polymerase	Cuts protein at specific sequence
c.	Helicase	Breaks the hydrogen bonds between complementary base pairs during DNA replication.

(1) a, b and c are correct

(2) a and b are correct but c is false

(3) a is correct but b and c are false

(4) a and c are correct but b is false

78) *E. coli* bacteria were grown on heavy nitrogen medium for several generation and then transferred to normal or light nitrogen source. What must be the number of pure heavy DNA molecules after four generations on light nitrogen medium ?

(1) Zero

(2) Two

(3) Four

(4) Eight

79) Which statement about nucleotides is correct ?

(1) Nucleotides such as ATP are used in the synthesis of nucleic acid.

(2) Nucleotide are composed of only a pentose sugar and a nitrogen bases.

(3) Nucleotide such as TTP used in the synthesis of RNA molecules.

(4) Nucleotide contain a deoxyribose sugar if they are components of RNA.

80) Against the codon 5'UCC3', what would be the sequence of anticodon on tRNA ?

(1) 5' AGG 3'

(2) 5' ATG 3'

(3) 5' GTA 3'

(4) 5' GGA 3'

81) **Assertion (A) :** The split gene arrangement represent probably an ancient feature of the genome.
Reason (R) : hnRNA undergoes additional processing called as capping and tailing.
Choose the correct option.

(1) (A) and (R) are correct, (R) is correct explanation of (A)

(2) (A) and (R) are correct, (R) is not correct explanation of (A)

(3) (A) is incorrect, (R) is correct

(4) Both (A) and (R) are incorrect

82) When an important predator starfish Pisaster was experimentally removed from an enclosed intertidal area of communities of the American pacific coast, it resulted in extinction of more than 10 species of invertebrates with in a year. It demonstrates role of a predator as a factor for :-

(1) preventing mutualism among prey species

(2) reducing the intensity of competition among competing prey species

(3) Over-exploitation of prey species

(4) reducing reproductive success in prey species

83) Read statements (A–D)
(A) Sparrow eating seeds is no less a predator.
(B) Predator helps in maintaining species diversity in a community
(C) *Calotropis* produces cardiac glycoside that's why no cattle grazing on this plant
(D) Monarch butterfly is highly distasteful to predator.
How many statements are correct ?

(1) Two

(2) Four

(3) One

(4) Three

84) An association in which one species is inhibited by toxic secretion of another species :-

(1) Competition

(2) Amensalism

(3) Parasitism

(4) Commensalism

85) The process of mineralisation by microorganisms helps in the release of :-

(1) Inorganic nutrients from humus

(2) Organic nutrients from humus

(3) Both organic and inorganic nutrients from detritus

(4) Inorganic nutrients from detritus and formation of humus

86) Given below are two statements :
Statement-I : Decomposition rate are slower if detritus is rich in lignin & chitin and quicker, if detritus is rich in nitrogen and water soluble substances like sugars.
Statement-II : Break down of complex organic matter into inorganic substances like CO_2, water and nutrients is called decomposition.
Choose the correct answer from options given below -

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

87) Shape of an ecological pyramid was inverted. It could belong to:-

(1) Pyramid & Number in Grassland

(2) Pyramid of Number in sea

(3) Pyramid & Biomass in Grassland

(4) Pyramid of Biomass in sea

88) **Assertion** : In-situ conservation maintains the natural flow of evolution.
Reason : In-situ conservation allows species to interact with their environment, adapt to changing conditions, and continue to evolve.

(1) Both assertion and reason are true but reason is not the correct explanation of assertion.

(2) Both assertion and reason are true and reason is the correct explanation of assertion.

(3) Assertion is true and reason is false.

(4) Assertion is false but reason is true.

89) Which statement is/are correct :-
(a) Nile perch introduced into lake victoria lead of extinction of cichlid fish
(b) The earth submit held in Johannesburg took measures for biodiversity conservation
(c) exploring molecular genetics and species level diversity for products of economic importance called molecular genetics
(d) Almost 40 percent forest have been lost in the tropics.

(1) Only a & b are correct

(2) Only b & c are correct

(3) Only c & d are correct

(4) Only a & d are correct

90) Which of the following is the most important cause driving animals and plants to extinction ?

(1) Habitat loss and Fragmentation

(2) Over-exploitation

(3) Alien species invasions

(4) Co-extinctions

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	3	1	1	1	2	1	3	3	3	2	1	3	1	3	1	3	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	3	1	2	4	1	4	2	1	1	2	2	2	1	1	2	3	4	4
Q.	41	42	43	44	45															
A.	3	1	4	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	1	2	3	3	2	3	1	4	3	3	4	2	1	4	4	3	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	3	3	2	4	3	4	3	3	2	3	4	3	4	3	4	2	3	1
Q.	86	87	88	89	90															
A.	2	4	2	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	1	2	2	1	3	3	3	2	2	1	1	4	2	2	2	3	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	3	2	4	1	4	4	3	2	1	2	3	2	1	2	4	4	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	2	3	1	4	4	1	1	2	2	3	1	2	2	2	3	3	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	2	1	3	2	1	1	3	3	3	4	2	4	1	3	4	1	1	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	2	2	1	1	4	2	4	1										

SOLUTIONS

PHYSICS

1) Maximum % Error = $\frac{\Delta M}{M} \times 100 + \frac{2\Delta L}{L} \times 100 + \frac{2\Delta T}{T} \times 100 + \frac{\Delta A}{A} \times 100$
= 1 + 4 + 1 + 1
= 7%

2) At maximum height v = u cos θ
$\frac{u}{3} = u \cos\theta \Rightarrow \cos\theta = \frac{1}{3}$
$R = \frac{u^2 \sin 2\theta}{g} = \frac{2u^2 \sin\theta \cos\theta}{g}$
$= \frac{2u^2}{g} \times \frac{2\sqrt{2}}{3} \times \frac{1}{3} = \frac{4\sqrt{2}u^2}{9g}$
⇒ n = 4
⇒ 5n = 20

3) At t_1, tangent having θ < 90°
 ⇒ a = +ve & v = +ve
 ⇒ $\vec{a} \cdot \vec{v}$ is positive
At t_2, a = 0 & v = +ve
At t_3, a = –ve & v + ve ⇒ $\vec{a} \cdot \vec{v} = -ve$
If angle between $\vec{v}$ & $\vec{a}$ is less than 90° then particle speeds up.

4)

Given $m_1 = m_2$ and $\frac{r_1}{r_2} = \frac{3}{4}$
As centripetal force $F = \frac{mv^2}{r}$
In order to have constant (same in this question) centripetal force
$F_1 = F_2$
$\frac{m_1 v_1^2}{r_1} = \frac{m_2 v_2^2}{r_2} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{r_1}{r_2}} = \frac{\sqrt{3}}{2}$

5) By work Energy theorem
$W_{gravity} + W_{friction} + W_{spring} = \Delta KE$
$\Rightarrow mgx\sin\theta - \mu mg\cos\theta\, x - \frac{1}{2}Kx^2 = 0$
$\frac{1}{2}kx^2 + \mu mgx\cos\theta = mgx\sin\theta$

6) Photo diode detect light when

$$\Delta Eg \leqslant \frac{hc}{\lambda}$$

$$\Delta Eg \leqslant \frac{12400}{6000}$$

$$\Delta Eg \leqslant 2.06\,eV$$

7) Finge width, $\beta = \frac{D\lambda}{d} = \frac{1.5 \times 589 \times 10^{-9}}{0.15 \times 10^{-3}}$

$= 5.9 \times 10^{-3}$ m

= 5.9 mm

8) $(\mu - 1)\,t = n\beta$

$$\frac{(\mu_1 - 1) \times 1.8 \times 10^{-5}}{(\mu_2 - 1) \times 3.6 \times 10^{-5}} = \frac{18\beta}{9\beta}$$

$$(\mu_1 - 1) = 4(\mu_2 - 1)$$

$$4\mu_2 - \mu_1 = 3$$

9)

At saturation current all emitted electrons reach at anode and when mono chromatic light is used in photoelectric effect then range of kinetic energy of emitted electrons is zero to k_{max}.

10) $\frac{1}{2}mv_1^2 = \frac{hc}{\lambda} - \phi_0$ work function negligible $\phi^0 = 0$

$$\frac{1}{2}mv_1^2 = \frac{hc}{\lambda}$$

$$\frac{1}{2}mv_2^2 = \frac{hc}{4\lambda}$$

$$v_2^2 = \frac{v_1^2}{4}$$

$$v_2 = \frac{v_1}{2}$$

11) COLM

$$\frac{m_1}{m_2} = \frac{v_2}{v_1}$$

$$\frac{m_1}{m_2} = \frac{27}{1}$$

$R = R_0A^{1/3}$

$$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}}$$

$$\frac{R_1}{R_2} = \left(\frac{27}{1}\right)^{\frac{1}{3}}$$

$R_1 : R_2 = 3 : 1$

12) $r_n = 5.3 \times 10^{-11}\frac{n^2}{z^2}$

$$r_3 = 5.3 \times 10^{-11} \times \frac{(3)^2}{(1)^2} m$$

$= 47.7 \times 10^{-11}$ m

13)

$N \times 2\pi r = L$

$$r \propto \frac{1}{N}$$

$$B = \frac{\mu_0 NI}{2r} \propto \frac{N}{r}$$

$$\boxed{B \propto N^2}$$

14) Given : $L = 2 \times 10^{-4}$ H

$R = 6.28\ \Omega$

$f = 10$ MHz $= 10^7$ Hz

Since quality factor,

$$Q = \omega^0 \frac{L}{R} = 2\pi f \frac{L}{R}$$

$$\therefore Q = 2\pi \times 10^7 \times \frac{2 \times 10^{-4}}{6.28}$$

$Q = 2 \times 10^3 = 2000$

15)

Perfect insulator ($\sigma = 0$) and perfect conductor ($\sigma = \infty$) But real or imperfect medium both. I_c & I_d can co exist.

16) For force maximum

$$\frac{dF}{dx} = 0$$

$$\frac{9\sqrt{3}Qq}{4\pi\epsilon_0} \frac{d}{dx}\left(\frac{x}{(3x^2 + a^2)^{3/2}}\right) = 0$$

$$x = \frac{a}{\sqrt{6}}$$

17) $v_A - v_B$ = work done by electric field on + 1 coul. charge from A to B = E Rθ

$\therefore v_B = v_A - E R\theta = v - E R\theta$

18)

$$\text{Slope} \rightarrow \frac{Q}{\Delta T} = \frac{2500}{100}$$

$$\frac{nC_P \Delta T}{\Delta T} = 25$$

$1 \times C_P = 25$

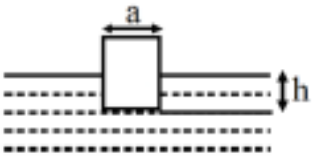
$$\frac{\gamma R}{\gamma - 1} = 25$$

$\frac{\gamma}{\gamma - 1} = 3 \Rightarrow \gamma = 3\gamma - 3 \Rightarrow 2\gamma = 3 \Rightarrow \gamma = 3/2$

19) $T_{mix} = \frac{m_1 s_1 T_1 + m_2 s_2 T_2}{m_1 s_1 + m_2 s_2} = \frac{2m\frac{C}{2}T + \frac{m}{3}2C\,3T}{2m\frac{C}{2} + \frac{m}{3}.2C}$

$T_{mix} = \frac{9T}{5}$

20)



Condition of equilibrium $mg = F_B$

$\Rightarrow mg = S_{liquid}\, a^2 hg$

When supressed by x net force $F'_B - mg = kx$

$S_{liquid}a^2(h+x)g - mg = kx$

$S_{liquid}a^2xg = kx$

$\Rightarrow k = S_{liquid}\, a^2 g$

$m = a^3\, s_{solid}$

$\therefore T = 2\pi\sqrt{\frac{m}{k}}$

$T = 2\pi\sqrt{\frac{a^3 S_{solid}}{S_{liguid}a^2 g}} = 2\pi\sqrt{\frac{as}{g}}$

21)

A. Initial time period: $T_1 = 2\pi\sqrt{\frac{L_1}{g}}$

B. New time period: $T_2 = 2\pi\sqrt{\frac{1.21L_1}{g}}$

$$T_2 = \sqrt{1.21} \times T_1 \approx 1.1T_1$$

Increase in time period:

$$\Delta T = 0.1T_1$$

The time period increases by 10%.

22)

$\Delta\ell = \alpha\, \Delta t.$

Strain = $\alpha\Delta t$

Stress = Y. Strain

= Y . α. Δt

T = (Area). (Stress) = (Area).YαΔt

$v = \sqrt{\frac{T}{\mu}}$

$f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = \frac{1}{2\ell}\sqrt{\frac{(\text{Area}).Y\alpha\Delta t}{\mu}}$

$= \frac{1}{2\times 1}\sqrt{\frac{10^{-6}\times 2\times 10^{11}\times 1.21\times 10^{-5}\times 20}{0.1}} = 11$

23)

$v = \pi r^2$

$\frac{\Delta v}{v} = \frac{2\Delta r}{r} + \frac{\Delta \ell}{\ell}$

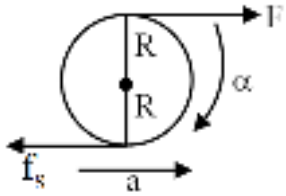
$\sigma = \frac{-\Delta r/r}{\Delta \ell/\ell} = \frac{1}{2}$

$-\frac{\Delta r}{r} = \frac{\Delta \ell}{2\ell}$

$\frac{2\Delta r}{r} + \frac{\Delta \ell}{\ell} = 0 \Rightarrow \frac{\Delta v}{v} = 0$

24) By perpendicular axis theorem moment of inertia about any axis passing through centre and in the plane of plate will be I (by symmetry)

25)



$F - f_S = ma$ ....(1)

$FR + f_S R = \frac{2}{5}mR^2\alpha$

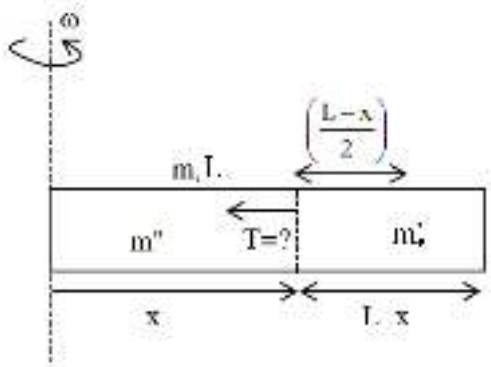
$\alpha = \frac{a}{R}$

$F + f_S = \frac{2}{5}ma$ ....(2)

Solving (1) and (2)

$f_S = -\frac{3F}{7}$ (Direction will be opposite to as shown)

26)



$T = m'\omega^2\left(x + \frac{L-x}{2}\right)$

$$T = \frac{m}{L}(L-x)\left(\frac{x+L}{2}\right)\omega^2$$

$$T = \frac{1}{2}\frac{m\omega^2}{L}(L^2-x^2)$$

Alternate Method,

Let assume mass of the small segment at a distance x is dm.

$dT = dm\omega^2 x$

$$dT = \frac{m}{L}.dx\omega^2 x$$

The tension T at distance x must provide the total centripetal force for all elements of the rod from x = x to the free end x = L.

$$\int_0^T dT = \int_x^L \frac{m}{L}\omega^2 x dx$$

$$T = \frac{m}{L}\omega^2\left[\frac{x^2}{x}\right]_x^L \Rightarrow T = \frac{m\omega^2}{2L}(L^2-x^2)$$

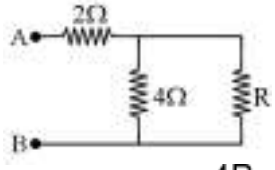
27)

$$v_d = \frac{I}{neA}$$

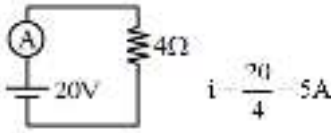
$$= \frac{10}{9\times10^{28}\times1.6\times10^{-19}\times1\times10^{-4}}$$

$\Rightarrow v_d = 6.94 \times 10^{-6}$ m/s

28) Let equivalent resistance Across A and B is R.



$$\text{then, } R = 2 + \frac{4R}{4+R}$$



$(4 + R)R = 8 + 2R + 4R$

$R^2 + 4R = 8 + 6R$

$R^2 - 2R - 8 = 0$

$R^2 - 4R + 2R - 8 = 0$

$(R - 4)(R + 2) = 0 \quad \Rightarrow \quad R = 4$

29)

For voltmeter

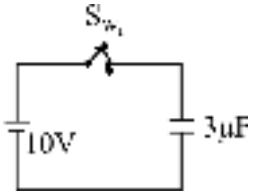
$(100 + R) \times 10^{-6} \times 50 = 50 \Rightarrow R \simeq 10^6\ \Omega$

For Ammeter

$100 \times 50 \times 10^{-6} = R \times 10 \times 10^{-3} \Rightarrow R = 0.5\ \Omega$

30)

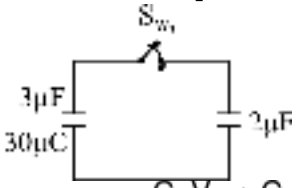
When S_{W_1} is closed & S_{W_2} is open



$Q_{3\mu F} = C \times V$

$= 3 \times 10 = 30\ \mu C$

When S_{W_1} is opened & S_{W_2} is closed :



So, $V_{cm} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2} = \frac{30 + 0}{3 + 2} = 6V$

Now, $Q'_{3\mu F} = C \times V_{cm} = 3 \times 6 = 18\mu C$

$Q_{2\mu F} = CV_{cm} = 2 \times 6 = 12\ \mu C$

31) Induced current in the circuit $i = \frac{Bvl}{R}$

Magnetic force acting on the wire $F_m = Bil = B\left(\frac{Bul}{R}\right)l$

$\Rightarrow F_m = \frac{B^2v l^2}{R}$. External force needed to move the rod with constant velocity.

$(F_m) = \frac{B^2vl^2}{R} = \frac{(0.15)^2 \times (2) \times (0.5)^2}{3}$

$= 3.75 \times 10^{-3}N$

32) For paramagnetic material

According to curies law

$\chi \propto \frac{1}{T}$

$\chi \propto \frac{1}{T} \quad \Rightarrow \quad \chi_1 T_1 = \chi_2 T_2$

$\Rightarrow \frac{6}{0.4} \times 4 = \frac{I}{0.3} \times 24$

$I = \frac{0.3}{0.4} = 0.75\ A/m$

33)

$u_{avg} = \varepsilon_0 E_{rms}^2$

$= 8.85 \times 10^{-12} \times 720 \times 720$

$= 4.58 \times 10^{-6}\ J/m^3$

34) After first refraction will be formed at V_1

$$\frac{\mu_2}{v_1} - \frac{\mu_2}{-\infty} = \frac{\mu_2 - \mu_1}{R} \quad \text{....(i)}$$

Image after second refraction will be found at V_2

$$\frac{\mu_3}{v_2} - \frac{\mu_2}{v_1} = \frac{\mu_3 - \mu_2}{R} \quad \text{....(ii)}$$

Adding (i) & (ii)

$$\frac{\mu_3}{v_2} = \frac{\mu_3 - \mu_1}{R} \quad \Rightarrow \quad v_2 = \frac{\mu_3 R}{\mu_3 - \mu_1}$$

$$\frac{\mu_3 R}{\mu_3 - \mu_1} = 2R$$

 focal length will be =

35) $f_0 = 2.5$ cm $\quad u_0 = -3.75$cm

$f_e = 5$ cm

$$v^0 = \frac{f_0 u_0}{u_0 + f_0} = \frac{(2.5)(-3.75)}{-1.25} = 7.5 \text{ cm}$$

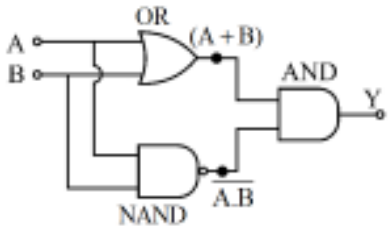
$$L_{min} = V_0 + \frac{f_e D}{f_e + D} = \left[7.5 + \frac{(5)(25)}{(5+25)}\right] \text{cm}$$

$L_{min} = 11.67$ cm

36) $$\frac{\sin\theta_a}{\sin\theta_b} = \frac{n_b}{n_a} = \frac{v_a}{v_b} = \frac{v_a}{c/3}$$

$$v_a = \frac{c}{3} \times \frac{\sin 45^\circ}{\sin 60^\circ} = \frac{c}{3}\sqrt{\frac{2}{3}}$$

37)



$Y = (A + B).(\overline{A.B})$

$Y = (A + B).(\overline{A} + \overline{B})$

$Y = A.\overline{A} + A.\overline{B} + B.\overline{A} + B.\overline{B}$

$Y = A.\overline{B} + B.\overline{A}$

38) $\because T^2 \propto V^3$

$$\left(\frac{PV}{nR}\right)^2 \propto V^3$$

$P^2V^{-1} = \text{const}$

$PV^{-1/2} = \text{const} \Rightarrow x = -1/2$

$$C = C_v + \frac{R}{1-x} = \frac{5R}{2} + \frac{2R}{3} = \frac{19R}{6}$$

39) $E = \sigma T^4$

$$R = \sigma\left(\frac{b}{\lambda_m}\right)^4$$

$$\left(\frac{R}{\sigma}\right)^{1/4} = \frac{b}{\lambda_m}$$

$$\lambda_m = b\left(\frac{\sigma}{R}\right)^{1/4}$$

40)

$$[K] = \frac{[V][Y_2]}{[x_2]^4} = \frac{[M^0L^3T^{-1}][L]}{[L^4]}$$

$$[K] = [M^0L^0T^{-1}]$$

41)

$$1 \text{ MSD} = \frac{1}{10} = 0.1 \text{ cm}$$

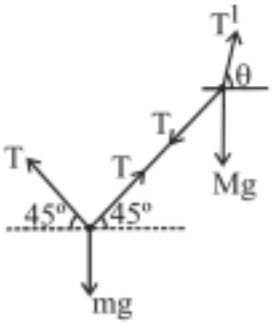
10 VSD = 8 MSD

1 VSD = 0.8 × 0.1 = 0.08 cm

L.C. = 1 MSD – 1VSD

= 0.1 – 0.08

= 0.02 cm



42)

$$\frac{T}{\sqrt{2}} + \frac{T}{\sqrt{2}} = mg$$

$$T = \frac{mg}{\sqrt{2}}$$

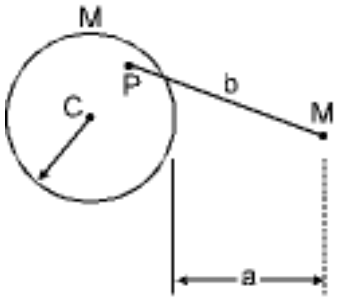
$$T' \cos\theta = \left(\frac{mg}{\sqrt{2}}\right)\frac{1}{\sqrt{2}}$$

$$T' \sin\theta = \frac{mg}{\sqrt{2}}\frac{1}{\sqrt{2}} + mg$$

$$\tan\theta = \frac{M + \frac{m}{2}}{\frac{m}{2}}$$

$$\tan\theta = 1 + \frac{2M}{m}$$

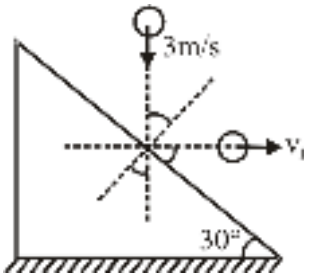
43) At centre



$V_c = -\frac{GM}{a} - \frac{GM}{2a}; \; E_c = \frac{GM}{(2a)^2}$

At any point P inside $V_P = -\frac{GM}{a} - \frac{GM}{b}$

$E_P = \frac{GM}{b^2}$ {only due to outside mass M}



44)

$3 \sin 30° = v_f \cos 30°$

$3\left(\frac{1}{2}\right) = v_f \frac{\sqrt{3}}{2}$

$v_f = \sqrt{3}$ m/s

45)

$V = \frac{2}{9\eta} r^2 (\rho - \sigma) g$

Here $v_1 = v_2$

$r_1^2(\rho_1 - \sigma) = r_2^2(\rho_2 - \sigma)$

$\frac{r_1^2}{r_2} = \frac{11-2}{8-2} = \frac{3}{2} \Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{3}{2}}$

CHEMISTRY

46) $\text{mole} = \frac{3.33}{666} = \frac{1}{2} \times 10^{-2}$

molecules of water = $18 \times \frac{1}{2} \times 10^{-2} \times 6.02 \times 10^{23} = 9.0 \times 6.02 \times 10^{21}$

47)

$\Delta x.\Delta v = \frac{h}{4\pi m}$

$\Delta x = \frac{h}{4\pi} \times \frac{1}{m} \times \frac{1}{\Delta v}$

$\Delta x = 0.527 \times 10^{-34} \times \frac{1}{9.1 \times 10^{-31}} \times \frac{1}{2.2 \times 10^{6}} \times \frac{100}{0.1}$

$\Delta x = \frac{0.527}{9.1 \times 2.2} \times 10^{-34+31+3-6}$

$\Delta x = \frac{527}{91 \times 22} \times 10^{-6} = 0.0263 \times 10^{-6}$ m

$= 26 \times 10^{-9}$M = 26 nm

48)

$E_2 = \frac{-13.6 \times 1^2}{2^2} =$ –328 ...(1)

$E_4 = \frac{-13.6 \times 1^2}{4^2} = x$ ...(2)

$\frac{E_2}{E_4} = \frac{16}{4} = \frac{-328}{x}$

$x = \frac{-328 \times 4}{16}$

x = –82 kJ/ mol

49) $(NH_4)_2CO_3(s) \rightleftharpoons 2NH_3(g) + CO_2(g) + H_2O(g)$

2p p p

$K_p = (2p)^2 \, p.p = 64$ atm

p = 2 atm

$p_{total} = 2p + p + p = 8$ atm

50)

Catalyst do not change position of equilibrium.

51)

In first reaction reactant present are in gaseous state so heat released will be higher but in 2nd reaction c_{gr} first converted to (g) and H_2 converted to $2H_{(g)}$ so heat released will be consumed and y < x.

52) $r = k[A]^x$ so on decreasing concentration rate decreases $(t_{1/2})^0 = \frac{A_0}{2k}$; where k is temperature dependent so $t_{1/2}$ is also temperature dependent.

53)

In SO_4^{2-} sulphur is present in highest oxidation number +6.

54) $\Delta G^o_{200} = \Delta H^o_{200} - T\Delta S^o_{200}$
$\Delta H^o_{200} = 20 - 4 = 16$ kJ/mol
$\Delta H^o_{T_2} = \Delta H^o_{T_1} + \Delta C_p[T_2 - T_1]$
$\Delta H^o_{400} = \Delta H^o_{200} + \frac{20 \times 200}{1000}$ kJ/mol
= 16 + 4 = 20 kJ/mole

55)

$\Delta G^0 = -nFE^0$
$= -2 \times 96500 \times (1.23 - 1 - 0.44)$ J
$\Delta G^0 = -2 \times 96500 \times 1.67 = -322000$ J
= – 322 KJ

56)

$$H_2O \rightleftharpoons H^+ + OH^-$$
t=0 C → 0 0
at equilibrium (C–Cα) → Cα Cα
$K = \frac{C\alpha . C\alpha}{C(1-\alpha)} = C\alpha^2 = 55.55 \times (18 \times 10^{-10})^2$
$= 55.55 \times 324 \times 10^{-20}$
$= 1.8 \times 10^{-16}$

57)

Given: pA° = 450 mm, pB° = 700 mm, P = 600 mm

xA = (P − pB°) / (pA° − pB°)
xA = (600 − 700) / (450 − 700) = (−100)/ (−250) = 0.4

xB = 1 − xA = 0.6

yA = (xA·pA°) / P = (0.4 × 450) / 600 = 180 / 600 = 0.3

yB = 1 − yA = 0.7
Correct statements: A, C and D

58) **Given Data :** m = b, $W_{solution}$: ag , W_{solute} = cg

Concept : $m = \frac{W_{solute} \times 1000}{(M_m)_{solute} \times W_{solvent(g)}}$

$W_{solvent} = W_{solution} - W_{solute} = (a - c)$ g

$b = \frac{c \times 1000}{(M_m)_{solute} \times (a-c)}$

$$\Rightarrow (M_m)_{solute} = \frac{c}{b} \times \frac{1000}{(a-c)}$$

59) $E^0_{cell} = (E^0_{RP})_C - (E^0_{RP})_A$

60) Fact

61)

(A) Lake Test → Al^{+3}
(B) Nessler's Reagent → NH_4^+
(C) Potassium sulphocyanide → Fe^{+3}
(D) Brown ring Test → NO_3^-

62)

B < Al ≥ Ga < In ≤ T□ (Atomic size)

63) Table 3.5 NCERT page No. 80

64) The bond order in diamond is 1. Graphite has bond order 1.33 and in fullerene bond has double character. So diamond has largest bond length.
Due to close packing of atoms in diamond, it has maximum density (3.52g/ml)
Since diamond and graphite have infinite 3D structure, while fullerenes are discrete molecule so dangling bonds are absent in fullerenes.

65) Unlike NH_3, PH_3 molecules are not associated through hydrogen bonding in liquid state. That is why the boiling point of PH_3 is lower than NH_3.

66)

Lanthanoid +3 oxidation state is more stable.

67)

(A) Fe: Most common stable states are +2 and +3. →(ii)
(B) V: Most common is +5 (in VO_2^+), with +4 and +3 also important. →(iv)
(C) Mn: Exhibits the maximum range; +2 is most stable, and the highest is +7. →(i)
(D) Th: Most common stable state is +4

68)

A. Coordination isomerism: interchange of ligands between cationic and anionic complexes.
B. Ambidentate ligand: can coordinate through two different donor atoms (e.g., SCN^-).

C. Ionisation isomerism: occurs when a counter ion is a potential ligand and exchanges places with a ligand.
D. $[Co(NH_3)_6][Co(CN)_6]$ → can exhibit coordination isomerism (interchange NH_3 and CN^- between cation and anion).

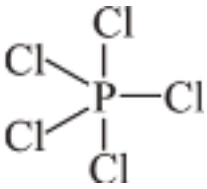
69)

Magnetic behaviour depends on the number of unpaired electrons and ligand field strength.
(1) Correct → For d^1–d^3 configurations, pairing not affected by ligand nature.
(2) Correct → d^4–d^7 complexes with strong field ligands are low spin.
(3) Incorrect → Magnetic moment depends on ligand nature (SFL vs WFL).
(4) Correct → CN^- causes pairing, Cl^- does not → different unpaired electrons.

70) (A) $[Pt(NH_3)_2Cl_2]$: Square planar → cis/trans geometrical isomerism → correct.
(B) Optical isomers: Non-superimposable mirror images → called enantiomers → correct.
(C) Chlorophyll: Contains magnesium in a porphyrin ring → correct.
(D) Ethylenediamine (en): Bidentate ligand, not monodentate → incorrect.

71) SiO_2 due to 3 dimensional giant network solid
CaO Ionic compound CCl_4 covalent molecule metallic-Zinc

72)



(a) False – structure is trigonal **bipyramidal**, not pyramidal.
(b) False – axial (90°) and equatorial (120°) bond angles differ.
(c) True – axial P–Cl bonds are longer due to greater repulsion.
(d) True – symmetrical structure → net dipole moment = 0.

73) (i) Group number of inert gas = zero group or18th group
(ii)

	ns	(n–2)f	(n–1)d	np
n = 6	6s	4f	5d	6p

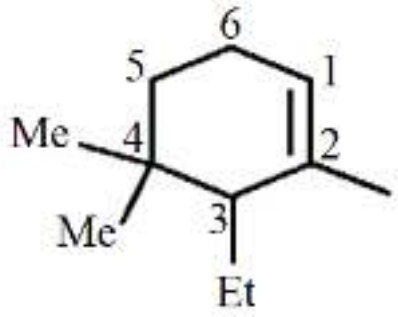
(iii) 2nd transition series = Y to Cd
(iv) Due to debye force

74)

Chlorous acid + Chloric acid
$HClO_2 + HClO_3$
↓ $-H_2O$
Cl_2O_4 or ClO_2 (anhydride)

75)

IP ⇒ N > O
(BO ∝ stability)
$N_2 > N_2^+ > N_2^-$ (stability)
$B_2 < C_2$ (stability)
Catenation
P – P > N – N
(More □.P. - □.P. repulsion)



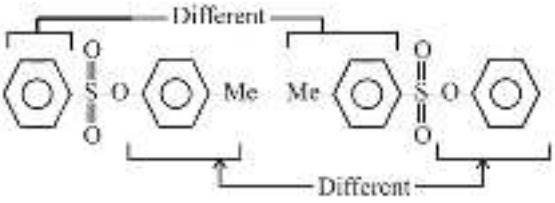
76)
3-Ethyl–4, 4–dimethyl cyclohex–1–ene

77)

Solution :-

Alcohols having $R-\underset{\displaystyle OH}{\underset{|}{C}}H-CH_3$ (group)
give iodoform test.
* If R = H, then alcohol will be primary (1°). Hence only ethanol is the primary alcohol that given iodoform test. Rest alcohols are secondary (2°) alcohols that can be give iodoform test.
Statement 1 is correct but statement 2 is wrong.
Final Answer :- (3)

78) Glucose is soluble in water due to presence of alcohol functional group and extensive hydration energy.



79)
Different group/substituent attached to polyvalent functional group :- Metamerism.

80)

Reagent used in clemenson reduction is Zn–Hg / conc. HCl

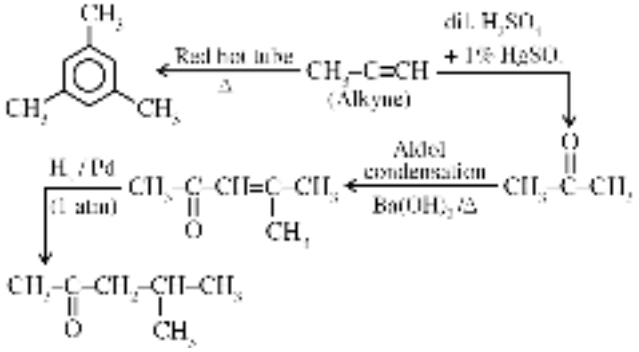
81) $Na_2S \quad Na_2[Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS]$ Violet Colour

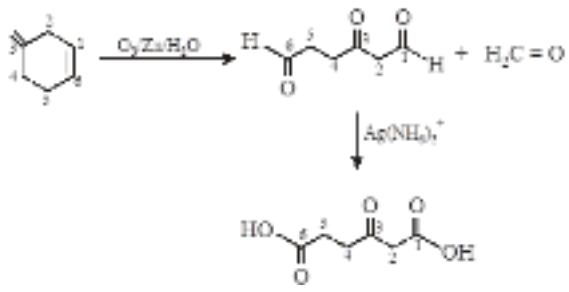
82)

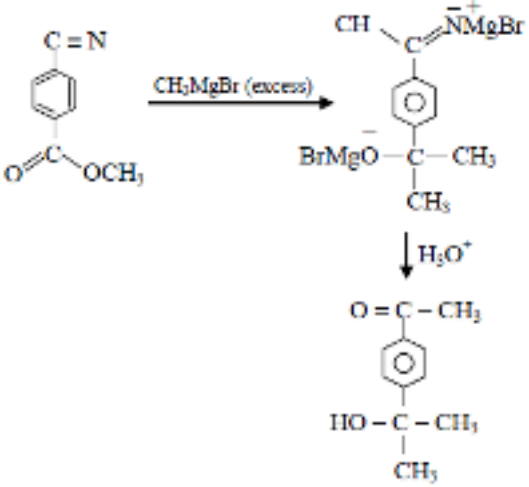
In S_N2 reaction
no carbocation is formed

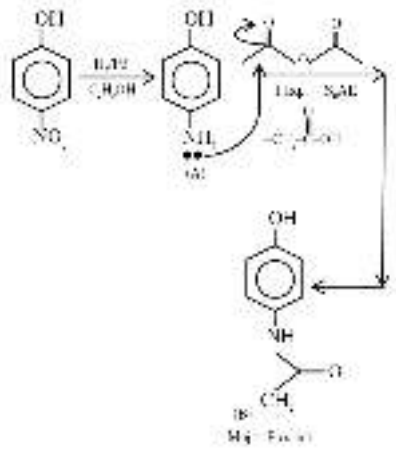
83) $R_f \text{ value} = \frac{\text{distance covered by component}}{\text{distance covered by solvent}}$

$= \frac{6}{12} = 0.5$

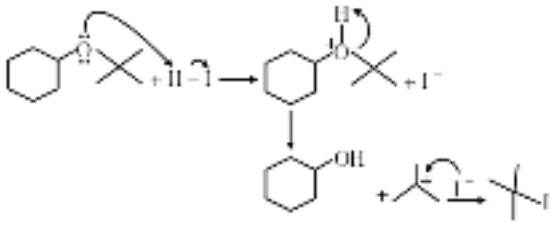
84) 

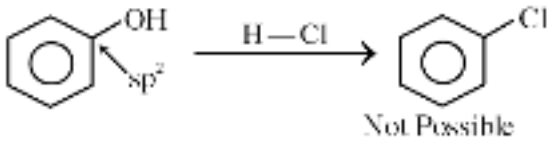
85) 

86) 

87) 

88) Due to H-bonding boiling point of alcohol is High.

89) 

90) 

BIOLOGY

91) NCERT, Pg. # 4

92)

NCERT Pg. 43, 44, 45, 48

93) NCERT, Pg. # 40,41,42

94) NCERT, Pg. # 45

95) NCERT, Pg. # 46

96) NCERT, Pg. # 49

97) Module

98) Module

99) NCERT XI (NEW) PAGE NO. 83

100) NCERT, Pg. # 203

101) NCERT (XII) ; Page No. # 196

102) NCERT, Pg. # 190

103) NCERT, Pg. # 189

104) NCERT, Pg. # 212

105) NCERT Pg.#209

106) NCERT, Pg. # 233

107) NCERT, Pg # 236

108) NCERT, Pg. # 242

109) NCERT, Pg. # 243

110) NCERT, Pg. # 239

111) NCERT, Pg. # 312

112) NCERT Pg. No. # 218

113) NCERT Pg. No. # 224

114) NCERT Pg. # 29

115) NCERT, Pg. # 31,32

116) NCERT-XI Pg. # 29,32,37

117) NCERT, Pg. # 38

118) NCERT, Pg. # 138

119) NCERT, Pg. # 130-136

120) NCERT, Pg. # 143

121) NCERT Pg. # 125

122) Reducing atmosphere

123)

NCERT Based

124)

NCERT Based

125) NCERT XII Page No. # 126 [New]

126) NCERT Pg. # 124

127)

NCERT XI Pg. # 126, 128, 131, A and D are correct

128) **Explain Question:** The question asked to evaluate the correctness of two statements:
Concept : This question is based on chromosomes
Explanation Statement I: During cell division, chromatin condenses into visible, structured chromosomes, and the nucleus disappears. This is correct.Statement II: Telocentric chromosomes have the centromere at the terminal end. This is also correct.
Final Answer: 4_Both Statement I and Statement II are correct

129)

NCERT–XI, Pg#135

130) NCERT Pg.#166,167.

131) NCERT Pg. # 180, 181

132)

NCERT Pg. # 180, 181, 184

133) NCERT Pg. # 167

134) NCERT, Pg # 169

135) NCERT Page No. 152, 153

136)

NCERT-XI _ Pg. # 11

137) NCERT-XI_Pg.# 17, 18

138)

NCERT-XI_ Pg. # 13, 27

139)

NCERT-XI _ Pg. # 29

140) NCERT XI_Pg. # 27

141)

NCERT-XI_pg. # 24

142) NCERT (XI) Pg. # 60,61

143) NCERT Pg.60

144) NCERT_Pg. # 62 and 63.

145) NCERT, Page No. 6,7

146) NCERT XI Pg. # 29, 30

147)

NCERT-XI, Pg. # 74

148) NCERT [XI] Pg.# 72

149)

NCERT-XI, Pg. # 110

150)

NCERT-XI_pg. No. 135

151)

NCERT XI Pg.# 114

152)

NCERT-XI_Pg. # 138

153) NCERT-XI Pg # 154

154) NCERT (XI)_Pg. # 159

155)

NCERT-XI_Pg. 176, 177

156)

NCERT-XI, Pg. # 174, 175, 176

157)

NCERT-XI_Pg. # 169

158)

NCERT-XII_Pg. # 7

159)

NCERT-XII_Pg. # 7

160) NCERT (XII) Pg. # 22,26

161)

NCERT-XII_ Pg. # 69

162)

NCERT XII Pg # 90

163)

NCERT Pg#85

164)

NCERT-XII, Pg. # 84

165) NCERT XII Pg.# 85,86

166)

NCERT-XII_ Pg. # 81
Question Explanation: Uracil is a nitrogenous base found in RNA**.**

Concept: Nitrogenous Bases

Solution: The molecule shown is Uracil. It is a nitrogenous base of the pyrimidine class. It is a component of RNA, where it pairs with adenine.

Final Answer: Option (3). 3

167)

NCERT-XII_ Pg. # 90
Question Explanation :Match enzymes of DNA replication with their correct function.
Concept:: **Molecular Basis of Inheritance: DNA Replication Enzymes**
Solution:

A. a. Ligase: Ligase is an enzyme that joins short segments of DNA together, such as during the repair or replication of DNA. This is correct.
B. b. DNA polymerase: DNA polymerase is responsible for adding nucleotides to a growing DNA strand during replication, not cutting proteins at specific sequences. This is incorrect.
C. c. Helicase: Helicase is an enzyme that breaks the hydrogen bonds between complementary base pairs during DNA replication, unwinding the DNA double helix. This is correct.

Final Answer : Option **(4)**

168) NCERT XII, Pg # 105

169) NCERT XII, Pg.# (E) 199; (H) 218

170) NCERT-XII_Pg # 98
Explain Question : The question asks to find the tRNA anticodon sequence complementary to the given mRNA codon 5' UCC 3'.

Concept : Genetic Code

Explanation :

mRNA codon = 5' U C C 3'

Complementary anticodon (3' → 5') = 3' A G G 5'

Writing anticodon 5' → 3' = 5' G G A 3'

Final answer : Option (4)

171) NCERT-XII Pg. # 95(E)/104(H)

172)

NCERT-XII_Pg. # 198
The question refers to the well-known experiment by Robert Paine in which the removal of the starfish Pisaster (a keystone predator) led to the extinction of several species in an intertidal community. This experiment demonstrates the important ecological role that predators can play in maintaining biodiversity.
Correct Answer 2. Reducing the intensity of competition among competing prey species
Explanation Correct. The removal of the starfish led to a dramatic increase in the population of mussels which dominated the space. This allowed them to outcompete other species for space and resources, resulting in the extinction of more than 10 species of invertebrates. The predator (Pisaster) helps keep dominant species in check, thus allowing other species to survive.

173)

NCERT Pg # 181

174) NCERT-XII_ Pg. # 197
Explain Question – The question asks: Which ecological interaction occurs when one species is inhibited by the toxic secretion of another species?
Concept – This question is based on Population interactions (amensalism and allelopathy)
Solution – In amensalism, one species is harmed while the other remains unaffected. A classic example is allelopathy, where plants like Parthenium or Black walnut release chemicals that

inhibit the growth of nearby plants.
NCERT Pg. # 197
Final Answer – Option 2

175) NCERT Pg. # 208

176) NCERT Page No. - 208

177) NCERT 12th, Page No. # 213

178) NCERT page No. - 224,225

179)

NCERT-XII_ Pg. # 222, 223, 225
Question Explanation : Identify which of the given statements are correct regarding environmental and Biological facts.

Concept : This question is based on Biodiversity and Conservation.

Solution :
Statement 'b' is incorrect because, The earth summit held in Rio-de-Janeiro not in Johannesburg took measures for Biodiversity conservation.
Statement 'c' is incorrect because exploring molecular genetics and species level diversity for products of economic importance called Bioprospecting not molecular genetics.
Final Answer : option (4) Only a & d are correct

180) NCERT XII Pg. No. # 222-223